



Inverted fee structures, tick size, and market quality[☆]

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ABSTRACT

Stock exchanges compete for order flow through their fee models. A traditional model pays rebates to liquidity suppliers, and an inverted model pays rebates to liquidity demanders. Using a regulatory intervention to examine the interaction between tick size, restrictions on dark trading, and exchange fees, we show that traders use inverted venues to adjust for suboptimal tick sizes. Increased inverted venue activity improves pricing efficiency and liquidity, especially when the tick size is binding. We show that the sub-tick price improvement offered by inverted venues enhances competition for liquidity provision and increases information impounded into prices through nonmarketable limit orders.

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1. Introduction

Fragmentation is a defining feature of modern markets. Stock exchanges compete for order flow on many dimensions, one of which is trading fees. The most common

model is the make-take fee model, with 8 of the 13 US exchanges using this model. Under the make-take model, liquidity demanders (takers) are charged a fee, while liquidity providers (makers) receive a rebate. More recently, three exchanges, BATS-Y, NASDAQ BX, and EDGA, have adopted an inverted fee model, offering rebates to liquidity demanders and charging fees to liquidity providers.¹

The inverted fee model, and rebates more generally, have been a controversial feature of the US market. Maker-taker fees and rebates alter the compensation received by and paid to liquidity suppliers and demanders. These incentives change traders preferences for making versus taking liquidity. The fee/rebate differences also influence where traders route their orders and alter the probability of execution of displayed orders. Nonmarketable limit orders displayed on inverted venues will likely execute before orders displayed at the same price on traditional

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¹ The fee models themselves are also extremely complex, with exchanges charging different fees depending on participant activity level and stock price. Fee changes are also frequent. (See RBC Capital Markets, 2016. Complexity of exchange pricing and corresponding challenges to transparency and routing.)

make-take venues due to the rebate paid to liquidity takers. However, inverted venues are also less likely to attract liquidity suppliers. These dynamics substantially increase the complexity of the trading process.

The complexity created by make-take fees is further compounded by the fact that fees render quotes less informative because they are displayed without adjusting for fees (Harris, 2013). Other concerns about these fee models include potential agency conflicts between brokers and clients where brokers charge a commission to their clients rather than passing on the fees/rebates (Battalio et al., 2016). They also encourage greater fragmentation, with each exchange operator creating multiple venues where the only substantial difference between venues is the fee structure (Spatt, 2016).

Despite these negatives, it is important to note that neither the inverted fee model, nor the make-take fee model emerge randomly. In fact, trading fees evolve as the competitive and regulatory landscape changes.² Angel et al. (2011) argue that the different fee structures enable sub-tick trading, which helps traders gain queue priority. Chao et al. (2017) highlight the importance of various fee models as a competitive tool for exchanges. Their model shows that one of the drivers of innovation in fee models is discrete tick sizes. They argue that exchanges use make-take fees to create sub-tick transaction prices to meet market participants' trading needs. This argument is also consistent with Foucault et al. (2013), who report that discrete tick sizes prevent prices from being adjusted in a manner to neutralize the impact of fees/rebates. However, there is little empirical evidence showing how the fee structure affects the way market participants adapt their trading strategies in response to the changing regulatory environment and how this, in turn, affects market quality.

In this paper, we examine the interaction between the tick size, restrictions on dark trading, and exchange fee structures.³ Our research unveils how market participants use exchanges with varying fee structures to adapt to regulatory changes. We focus on changes resulting from an experiment implemented by the Financial Industry Regulatory Authority (FINRA) and the Securities and Exchange Commission (SEC), known as the Tick Size Pilot (the pilot).⁴ The pilot increased the minimum tick size from \$0.01 to \$0.05 for three groups of pilot stocks. Controversially,

one group of pilot stocks is also subject to a “trade-at” rule. This rule prohibits price matching unless an exception applies. The trade-at rule effectively shuts down the opportunity to trade at the best displayed quote or in the dark at sub-tick increments other than the midpoint. This rule means that trading in the dark requires price improvement of at least half a tick size.⁵ This makes the finer pricing grid offered by inverted fee venues more attractive to liquidity suppliers. The average market share of inverted venues doubled from 9% to almost 20% for stocks subject to the trade-at rule. The surge of inverted trading provides a precious opportunity to examine the importance of sub-tick trading on inverted venues for market participants, as normally inverted fee venues account for only a small proportion of trades. The experimental design of the pilot allows us to assess the impact on market quality of market participants adjusting to regulatory constraints by making greater use of venues offering an inverted fee structure.

We begin by examining the effect of changes in inverted trading on pricing efficiency and liquidity. To understand how the inverted fee model might impact pricing efficiency, consider the following example. Suppose a trader receives information that alters her valuation so that it lies between the prevailing best quotes and the next tick. It is challenging for the trader to profit from this information when depth on the traditional make-take venue is considerable, as her order will sit at the end of the queue with a low probability of execution. To increase this probability, she must provide price improvement by at least one tick, but her valuation implies this trade will not be profitable. Consequently, she does not trade and her information is not incorporated into prices. This problem is exacerbated when ticks widen, especially when the tick becomes larger than the existing bid-ask spread, and dark trading is restricted. The presence of an inverted fee exchange can change this outcome by allowing the trader to provide liquidity within the spread. Because the regulator enforces price priority across venues, but not time priority, the inverted venue can attract marketable orders with rebates. This gives the trader's nonmarketable limit order a higher probability of execution by paying a small execution fee. If the fee is smaller than the value of her information, the trader can profit, and the information is incorporated into prices. In short, the inverted fee structure allows sub-tick trades to capture valuations that lie within one tick, thereby improving pricing efficiency.

Our empirical results support this story. Increases in inverted trading resulting from the trade-at rule improve pricing efficiency. This result is driven by stocks for which the spread is constrained by the minimum tick size and is not present in less constrained stocks. This suggests that the sub-tick pricing grid offered by inverted venues is most valuable when the tick size is too large, which inhibits traders from exploiting their information. While inverted trading induced by the trade-at rule is beneficial, we find that the rule itself has little impact on pricing efficiency

² Historically, exchanges charged fees to both demanders and suppliers of liquidity, but in the late 1990s, Island ECN introduced the concept of a liquidity supplier rebate in an attempt to incentivize liquidity provision on its new trading venue. Reg NMS, which was introduced in 2005 to regulate competition among exchanges, prohibits from competing on tick size, but not on fee structures.

³ Dark trading is trading that occurs without pretrade transparency. In the US, dark trading takes three forms: (i) hidden orders on exchange order books; (ii) dark pools operated by brokers and proprietary trading firms, which are regulated as alternative trading systems (ATS); and (iii) internalization of order flow by broker-dealers. For further details about dark trading and its impact on market quality, see Comerton-Forde and Putnigš (2015) and Buti et al. (2011).

⁴ Rindi and Werner (2017) present a comprehensive overview of the impact of the pilot on liquidity and market activity. Albuquerque et al. (2017) find that liquidity and pricing efficiency decreases for treated stocks. Lin et al. (2017) show an increase in trading on inverted venues and dark venues following the pilot.

⁵ Comerton-Forde et al. (2018) examine the impact of the introduction of a trade-at rule in the Canadian market. They show that this rule eliminated the intermediation of retail orders in the dark and shifted retail orders onto the lit market with the lowest trading fee.

because it limits trading on dark venues, which also enables sub-tick trading. The benefits to pricing efficiency of trading on inverted venues are likely the result of a substitution effect, where inverted venues replace dark venues as the preferred platform for liquidity competition using sub-tick trading.

We also look at the impact of inverted trading on liquidity and find that the increase in inverted trading has a positive impact on liquidity. This result holds across both high and low tick constrained stocks. In contrast, the trade-at rule has a small and mostly negative impact on liquidity. Using granular stock-venue-level data to study order activity on inverted fee venues, we find that the trade-at rule increases the fraction of time inverted venues quote at the NBBO. This suggests that liquidity providers are more willing to offer sub-tick price improvement on inverted venues when the opportunity to do so in the dark is switched off. We also find an increase in the fraction of executions on inverted venues, conditional on one inverted venue being at the NBBO. This increases incentives to provide liquidity on inverted venues. Our results show that inverted venues enable market participants to adapt to the regulatory change and hence enhance competition among liquidity providers. The competition for sub-tick liquidity provision reduces trading costs and improves liquidity, which is consistent with the theoretical models of [Foucault and Menkveld \(2008\)](#) and [Werner et al. \(2015\)](#). Our results on the net impact of the pilot rules are consistent with [Rindi and Werner \(2017\)](#), who study the pilot and find larger quotes and lower volatility after the implementation of the larger tick sizes.

Inverted venues also experience more order activity after the introduction of the trade-at rule. The trade-to-order ratio, which is a proxy for execution probability, significantly increases much more for inverted venues than for other lit venues. This difference is stronger for highly tick constrained stocks, providing further evidence that sub-tick trading is especially valuable when the tick size is binding. To explore the economic implications of the change in order activity on inverted venues, we examine the price discovery on one representative inverted fee venue, NASDAQ BX, and its traditional fee counterpart, NASDAQ. We find that both marketable and nonmarketable limit orders on NASDAQ BX contribute more to the price discovery process after the implementation of the trade-at rule. Although we find that orders on NASDAQ also become more informative, the increase is smaller than for orders on NASDAQ BX. Our results for nonmarketable limit orders on NASDAQ BX are stronger in highly tick constrained stocks.

Consistent with orders on NASDAQ BX becoming more informative, we find that short selling, which is often associated with informed trading (e.g., [Boehmer et al., 2008](#); [Comerton-Forde et al., 2016](#)), increases on NASDAQ BX after the trade-at rule is implemented. Furthermore, we find that short selling by liquidity providers drives the increase in short selling on NASDAQ BX for tick constrained stocks. Our analysis of price discovery and short selling on NASDAQ BX further supports that inverted venues, with their sub-tick pricing grid, contribute to market quality by allowing market participants to adapt to regulatory con-

straints. Inverted venues are beneficial in incorporating new information into prices, especially when stocks are tick constrained.

Although our results indicate that market quality improves when participants adapt to the tick size pilot by routing more orders to inverted fee venues, we do not claim that the inverted fee structure is unconditionally optimal. We investigate only one channel through which inverted fee venues can affect market quality, namely by giving access to a preferential queue for a fee, and show that inverted venues can be beneficial when the tick size is binding.

Our paper relates most closely to the literature on sub-tick trading in dark pools and its impact on market quality. [Buti et al. \(2015\)](#) and [Kwan et al. \(2015\)](#) find that sub-tick trading is higher in liquid stocks and stocks with a high tick to price ratio (i.e., low priced), suggesting that sub-tick trading is driven at least in part by suboptimal tick sizes. [Buti et al. \(2015\)](#) also show that sub-tick trading improves market quality for liquid, high-priced stocks, but reduces it for illiquid, low-priced stocks. They also illustrate that its effects are dampened when the relative tick size is reduced on the lit market. Our results complement their findings for sub-tick trading on dark venues with results for sub-tick trading on lit venues.

We also build on the existing empirical work on exchange fees. [Cardella et al. \(2015\)](#) show empirically that fees are an important determinant of market share. They report that the total exchange fee relative to competitors is what matters and that the impact of take fees is stronger than make fees. Using an exogenous fee change, [Malinova and Park \(2015\)](#) show that make-take fees impact bid-ask spreads, the ratio of marketable to nonmarketable limit orders and execution rates. Our contribution is to consider the impact of inverted fee models rather than traditional make-take fee models.

Our findings also complement recent work by [Chao et al. \(2017, 2018\)](#) that sits at the intersection of these two strands of literature. They argue that discrete tick sizes encourage the use of make-take fee models and contribute to market fragmentation. Their results suggest that the wider tick size adopted in the pilot will increase fee dispersion and lead to greater fragmentation. Unsurprisingly, given that only a limited number of small firms are affected, exchanges have not adopted new fee models in response to the pilot, but we find evidence to support [Chao et al. \(2017\)](#)'s results on fragmentation.

2. Institutional details

Trading in US-listed equities is governed by Regulation National Market System (Reg NMS), which led to an increase in competition after it was implemented in 2005. Competition for trading services is intense, with activity split across 13 exchanges, approximately 40 ATS and over 200 broker-dealer internalizers. According to [Rosenblatt Securities Inc.](#),⁶ exchanges account for roughly 64% of consolidated volume, with no single exchange accounting for

⁶ [Rosenblatt Securities Inc.](#), 2016. Let there be light. US dark liquidity tracker, October.

more than 15% of consolidated volume. ATS, mainly in the form of dark pools, account for approximately 15% of consolidated volume. Broker-dealer internalizers account for the remaining 21% of consolidated volume.

The market design choices made by regulators in the implementation of Reg NMS fundamentally changed the trading landscape in the US equities market. Two elements of Reg NMS are particularly relevant for our paper: the order protection rule and the minimum price increment rule (minimum tick), as they constrain the way in which exchanges can compete against each other and against nonexchange trading venues.

Rule 611, the Order Protection Rule (OPR), requires exchanges to ensure that orders at the top-of-book at each exchange are filled before execution can occur at inferior prices on another exchange. The OPR does not extend to orders away from the top of the book and does not enforce time priority across venues. This rule requires participants to connect to all exchanges and has encouraged a proliferation of exchanges. Because this rule does not enforce time priority across venues, exchange operators have the incentive to create more limit order queues to give traders more opportunities to be at the front of the queue at any given price point. It also encourages order splitting across venues (Spatt, 2016), as traders use smart order routers (SORs) to direct order flow to both exchange and nonexchange venues to achieve best execution.

Rule 612 requires that the minimum tick size is \$0.01 for all stocks priced above \$1 and \$0.0001 for stocks priced below \$1. This rule encourages competition between traders based on speed, especially for stocks where the minimum tick size is binding (Yao and Ye, 2018). Although Rule 612 applies to both exchanges and nonexchange venues, the rule only prohibits venues from displaying, ranking, or accepting orders or indications of interests in sub-tick prices. It does not prohibit sub-tick trading. As a result, nonexchange venues, namely dark pools and wholesale internalizers of retail order flow, have been very successful at attracting order flow by offering price improvement within the National Best Bid and Offer (NBBO). Sub-tick trading is particularly high in stocks that are constrained by the minimum tick size. Kwan et al. (2015) argue that this rule also places exchanges at a competitive disadvantage relative to nonexchange venues.

Two other elements of market structure that are critical for our paper are the exchange fee models and the tick size pilot. Therefore we also explain these institutional details.

2.1. Exchange fee models

There are currently three exchange fee models offered by US equities exchanges. These are (i) a traditional make-take fee model where liquidity providers are paid a rebate and liquidity demanders are charged a fee, (ii) an inverted fee model where liquidity demanders are paid a rebate and liquidity suppliers are charged a fee, and (iii) a fee-only model where no rebates are paid. Eight of the 13 exchanges offer a traditional make-take fee model; three offer an inverted fee model, and two offer a fee-only model. Fig. 1 provides details of the fee model offered by each

exchange for Tape C securities (NASDAQ-listed securities). The fee models are similar for the other tapes.⁷

Rule 610 imposes an access fee cap of \$0.003. Given that exchanges typically use fees collected for trading to pay trading rebates, the access fee cap implicitly sets the maximum amount payable for rebates.⁸ It is clear from Fig. 1 that exchanges offering a traditional make-take fee model set the take fee at or close to the access fee cap. Some market participants argue that the exchange access fee creates an unlevel playing field between exchange and nonexchange venues, as it encourages traders to shift their activity to nonexchange venues. Fig. 1 also shows that the magnitude of the fees/rebates used by inverted venues is substantially lower. There is considerably more variation in the take rebate than in the take fee charged by the traditional fee venues.

An important feature of inverted venues is their ability to offer a finer pricing grid within the spread. To understand this feature, consider, for example, a stock listed on NASDAQ that currently has a best bid and ask of \$16.00 and \$16.01, respectively. Using the fees/rebates reported in Fig. 1, we observe that a liquidity provider that buys one share on NASDAQ, which uses a traditional make-take fee model, will pay the bid of \$16.00 and receive a rebate of \$0.0015, for a net purchase price of \$15.9985 that is outside the quoted spread. Similarly, a liquidity demander that buys one share at the ask of \$16.01 will pay a fee of \$0.003, for a net purchase price of \$16.013 that is also outside the quoted spread. In contrast, consider the same two trades being executed on NASDAQ BX, an inverted fee venue. The liquidity provider would pay a fee of \$0.002 for a net purchase price of \$16.002, while the liquidity taker would receive a rebate of \$0.0006 for a net purchase price of \$16.0094. Both net purchase prices fall within the quoted spread, and this result holds true for sells as well. The inverted fee model, therefore, enables traders to offer or receive sub-tick price improvement relative to the current NBBO.

2.2. Tick size pilot program

On August 24, 2014, the SEC announced a joint proposal with FINRA to hold a tick size pilot program. The pilot, which was approved on May 6, 2015, as a result of the 2012 Jumpstart Our Business Startups Act ("JOBS Act"), aims to "study the effect of tick size on liquidity and trading of small capitalization stocks." The pilot rules progressively came to effect over a transition period spanning the month of October 2016.

For the duration of the pilot, each eligible stock is assigned to one of three pilot test groups or the control group following a stratified sampling procedure. Each pilot test group consists of 400 stocks, and stocks within each group are subject to different rules. Pilot stocks in the control group may be quoted and traded at any trading increment currently allowed and serve as a baseline

⁷ Base fees for the different tapes are presented in Table A.1 of the Online Appendix.

⁸ Securities and Exchange Commission, 2015. Maker-taker fees on equities exchanges.

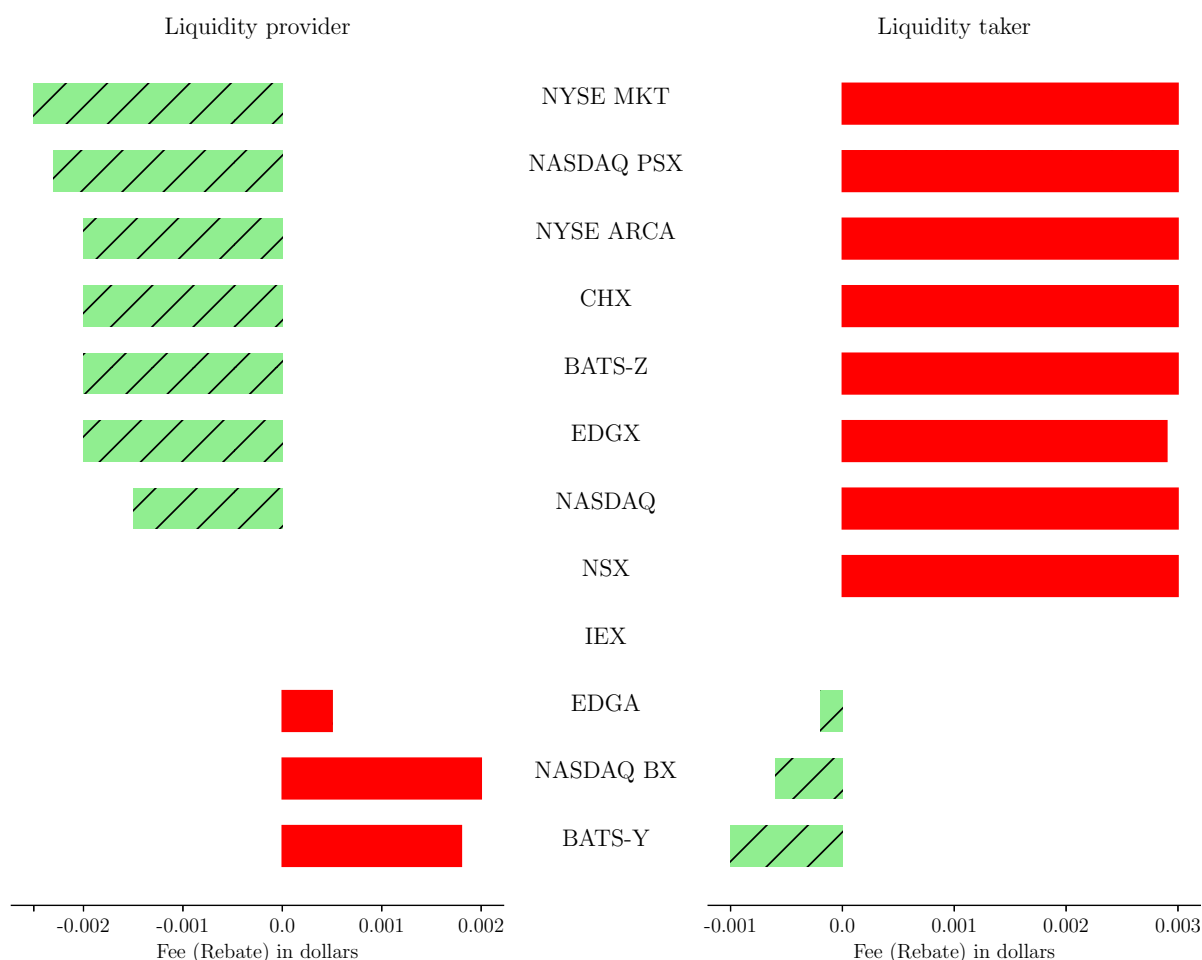


Fig. 1. Fees and rebates for Tape C stocks. This figure presents the standard exchange fee and rebate schedule, in \$ per share, for Tape C stocks (NASDAQ-listed stocks) priced above \$1 as at 21 December 2016. Positive numbers (red solid bars) indicate a fee while negative numbers (green hatched bars) indicate a rebate. The three venues with an inverted fee structure are BATS-Y, NASDAQ BX, and EDGA. Data are from each venue's respective website. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for the experiment. Pilot stocks in test group one are subject to a quoting requirement (quote rule) such that traders can only submit orders in \$0.05 increments, except for orders priced to trade at the NBBO midpoint or entered in a retail liquidity program. Pilot stocks in test group two are subject to the same quoting requirement as test group one but are also subject to a trading requirement (trade rule) that prohibits trades at price increments other than \$0.05, except for trades at the NBBO midpoint, retail investor orders with a price improvement of at least \$0.005, and negotiated trades.⁹ Finally, pilot stocks in test group three are subject to a “trade-at” prohibition (trade-at rule) in addition to the quoting and trading requirements. Under this trade-at rule, trading centers cannot execute orders at a protected price level for a quantity larger than

their displayed liquidity. In effect, this rule, combined with the trading requirements, prohibits dark venues from executing orders at prices other than the NBBO midpoint or with full-tick price improvement.¹⁰

The pilot came into effect following a staggered implementation schedule. The pilot began on October 3 with the first five stocks of test groups one and two. Ninety-five additional stocks were added to pilot groups one and two on October 10. The remaining stocks for groups one and two were activated on October 17. On the same date, the first five securities were added to pilot group three. Ninety-five additional stocks were added on October 24, and all remaining group three stocks were added on October 31.

Eligible stocks are those listed on NYSE, NYSE MKT, or NASDAQ that have a market capitalization of \$3 billion or less, a stock price of at least \$2.00, and an average daily volume of one million shares or less. Each stock is assigned

⁹ This rule is directed at broker internalization systems and dark venues. In the absence of this trading requirement, brokers are allowed to internalize client trades at a price improvement, and dark venues can accept quotes that offer price improvement as long as that price improvement is not guaranteed.

¹⁰ There are 13 exceptions to the trade-at rule. The main ones are for block trades and retail trades that offer at least \$0.005 of price improvement. See Section 2 of the Online Appendix for the full list.

Table 1

Descriptive statistics. This table presents the number of observations, mean, median, standard deviation, minimum, and maximum values and the 25th and 75th percentiles for trade volume market share and stock characteristics during the prepilot period of September 2016. All variables are daily stock-level observations, except for static controls for which there is one observation per sample stock. The sample includes all stocks from tick pilot groups and the control group. See the text for details on variable construction.

	N	Mean	Std	Min	25 Pct	Median	75 Pct	Max
Market share								
Dark market share (%)	47,711	38.88	17.82	0.01	26.01	35.76	48.78	100.00
Inverted market share (%)	47,711	8.63	5.84	0.00	4.30	8.43	12.16	62.52
Share of dark trading at mid-quote (%)	47,711	12.61	12.72	0.00	3.17	9.51	17.95	100.00
Share of dark trading at NBBO and price improving (non-MQ) (%)	47,711	80.20	17.89	0.00	72.20	83.86	93.22	100.00
Fragmentation ($\times 100$)	47,579	45.59	14.16	0.00	36.89	44.96	53.83	100.00
Stock characteristics								
Trade volume (\$ 1000 ^a)	47,711	3,393.92	7,330.40	0.00	157.33	1,004.85	3,825.55	344,527.39
Lit trade count	47,711	268.82	363.47	0.00	28.00	146.00	366.00	6,101.00
Time-weighted quoted spread (\$)	47,711	0.14	0.33	0.01	0.03	0.05	0.13	3.53
Amihud ratio ($\times 10^5$)	47,516	2.04	6.98	0.01	0.08	0.28	1.10	70.76
Log(Size)	2253	19.72	1.27	15.30	18.81	19.84	20.77	21.97
Stock price	2253	23.18	27.60	2.00	7.74	16.10	29.50	490.35
Listing exchange								
NYSE (%)	2253	29.47						
NYSE MKT (%)	2253	4.53						
NASDAQ (%)	2253	66.00						
Index membership								
Russell 1000 (%)	2253	1.69						
Russell 2000 (%)	2253	66.09						
S&P MidCap 400 (%)	2253	3.82						
S&P SmallCap 600 (%)	2253	22.41						
S&P Total Market (%)	2253	85.31						

to one of the test groups or the control group following a stratified sampling procedure.¹¹

3. Data description and sample statistics

Our sample comprises 2399 eligible pilot and control stocks. For each of these stocks, we obtain intraday trade and quote data for the consolidated tape and individual venue feeds from Thomson Reuters Tick History (TRTH) as provided by the Securities Industry Research Centre of Asia-Pacific (SIRCA). Our sample starts on September 1, one month before the pilot began, and ends on December 13, 2016, the day before the US Federal Reserve announced an increase of the Fed fund rate target range. We obtain market capitalization, share price and industry classification as of the end of August 2016 from the Center for Research in Security Prices (CRSP), and exchange-traded funds (ETF) holdings from the iShares website as of the end of August 2016 that we use as a proxy for index membership. We exclude stocks that are delisted or excluded from the pilot before the end of our sample and exclude November 25 because it is not a full trading day. Our final sample contains 385 stocks in group one, 386 in group two, 388 in group three, and 1170 in the control group, for a total of 2329 stocks, or 155,551 stock-day observations.

Table 1 reports summary statistics for the prepilot period, which we define as September 2016. For the typi-

cal stock in our sample, the median volume on inverted venues is 8.43% of the total volume, and the median volume on dark venues is 35.76%; 9.51% of the dark volume occurs at the midpoint of the NBBO, and the remaining is at the quote or price improving (i.e., executed at or within the NBBO spread but not at midpoint). Median fragmentation, measured as the inverse Herfindahl index of lit trading value during continuous trading, is 0.450. The median stock in our sample has a market capitalization of \$409 million, daily trade volume of \$1 million, and quoted spread of \$0.05. Approximately 66% of the stocks are included in the Russell 2000 index, and 2% are included in the Russell 1000.

We next assess the factors associated with the cross-sectional dispersion in the market shares of inverted and dark venues during the prepilot period. Table 2 presents coefficient estimates from ordinary least squares (OLS) regressions of the average daily market share of inverted and dark trading, respectively, during the prepilot period.

We control for a set of variables known to affect market fragmentation. Those control variables include the Amihud ratio, total daily volume, inverse of the price, and market capitalization and are also used by past studies as determinants for market fragmentation (see O'Hara and Ye, 2011; Buti et al., 2015). In general, we find that stocks with higher volume, better liquidity, lower prices, tighter spreads, and larger market capitalization have more inverted trading. While stocks with lower prices also have more dark trading, it is stocks with smaller market capitalization that attract more dark trading.

¹¹ For a technical description of eligibility criteria and the stratified sampling procedure, see Securities and Exchange Commission, 2015. Plan to implement a tick size pilot program. See Section 3 of the Online Appendix for an analysis of the sampling outcome.

Table 2

Determinants of market share. This table reports coefficients from OLS regressions of inverted venues and dark venues market share of dollar-volume continuous trading, both in %, on the Amihud illiquidity ratio (multiplied by a factor of 10^5), the log of the average daily dollar-volume ($\text{Log}(\text{Daily Volume})$), the inverse of the share price ($1/\text{Price}$), the log of the stock issue market cap ($\text{Log}(\text{Size})$), index membership fixed effects, and industry fixed effects. Both dependent variables, the Amihud illiquidity ratio, and the daily volume are measured daily and averaged over the pre-pilot period of September 2016. The share price and market cap are measured using the last closing price of August 2016. *t*-statistics are presented in parenthesis, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample includes all stocks from tick pilot groups and the control group. Sample descriptive statistics for the pre-pilot period are presented in Table 1.

	Inverted share		Dark share	
	(1)	(2)	(3)	(4)
Amihud ($\times 10^5$)	-0.028*** (-3.78)		-0.021 (-1.11)	
Log(Daily volume)		0.380*** (8.94)		-0.159 (-1.43)
1/Price	1.533* (1.87)	1.435* (1.79)	15.161*** (7.17)	14.661*** (7.00)
Spread	-0.889*** (-5.04)	-0.630*** (-3.57)	0.635 (1.40)	0.387 (0.84)
Log(Size)	1.444*** (18.81)	1.059*** (11.88)	-3.240*** (-16.39)	-3.014*** (-12.95)
Index F.E.	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes
Adjusted R^2	0.594	0.605	0.660	0.661
N	2329	2329	2329	2329

4. Distribution of trading volume across venues

We begin our analysis by examining how the distribution of trading volume across venues is affected by the pilot. Fig. 2 presents the average daily dollar volume for each tick pilot group and the control group. Vertical lines indicate dates when additional stocks are added to the pilot groups. While the overall trade volume fluctuates through time, no across-group difference emerges during the pilot. The pilot does not appear to affect volume, although volume is elevated in the week following the US presidential election (indicated by the dark shaded band).

Panels A and B of Fig. 3 present the daily mean share of trading volume for inverted and dark venues, for each tick pilot group and the control group, from September 1 to December 13, 2016. We define the daily share of inverted (dark) trading as the total dollar volume of continuous trading on inverted (dark) venues divided by the total dollar volume of continuous trading across all venues.

The pilot groups diverge from the control group in the pilot period. Panel A of Fig. 3 shows that more trading occurs on inverted venues in the pilot period. The increase is more important for stocks in group three than for stocks in groups one and two. The larger increase in group three stocks is an unintended outcome likely due to the inability to trade in the dark other than at the midpoint. In Panel B, we find that both group one and two experience an increase of approximately 5% in their dark market share. This is consistent with Kwan et al. (2015) and Buti et al. (2015) since an increase in the tick size pushes liquidity seekers onto dark venues to look for better exe-

cution prices within the spread. Unsurprisingly, dark trading decreases by over 5% for stocks in group three after the imposition of the trade-at rule. Panel C and D show that within dark trading, midpoint trading for group three stocks increases in the pilot period, whereas dark trading at NBBO or price improving over NBBO decreases.

The above graphical observation implies that the exogenously implemented tick pilot exerts a significant impact on inverted market share and dark share. We formally test these insights with the following OLS regressions:

$$\begin{aligned} \text{Inverted share}_{i,\tau} = & \alpha + \beta_1 \mathbf{1}_{[\text{trade-at}]_{i,\tau}} + \beta_2 \mathbf{1}_{[\text{trade}]_{i,\tau}} \\ & + \beta_3 \mathbf{1}_{[\text{quote}]_{i,\tau}} + \beta_4 \text{Inverted share}_{i,\tau-1} \\ & + \gamma X_{i,\tau} + \epsilon_{i,\tau} \end{aligned} \quad (1a)$$

$$\begin{aligned} \text{Dark share}_{i,\tau} = & \alpha + \beta_1 \mathbf{1}_{[\text{trade-at}]_{i,\tau}} + \beta_2 \mathbf{1}_{[\text{trade}]_{i,\tau}} + \beta_3 \mathbf{1}_{[\text{quote}]_{i,\tau}} \\ & + \beta_4 \text{Dark share}_{i,\tau-1} + \gamma X_{i,\tau} + \epsilon_{i,\tau}, \end{aligned} \quad (1b)$$

where $\mathbf{1}_{[\text{trade-at}]_{i,\tau}}$, $\mathbf{1}_{[\text{trade}]_{i,\tau}}$, and $\mathbf{1}_{[\text{quote}]_{i,\tau}}$ are dummy variables for the pilot rules. Dummies are equal to one if the rule is in effect for stock *i* at time τ , and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. *X* are controls including the daily CBOE Volatility Index (VIX) level, the log market cap ($\text{Log}(\text{Size})$), pilot group fixed effects, index membership fixed effects, industry fixed effects, and date fixed effects.¹² Standard errors are clustered by date and stock in the regression analysis.

Table 3 presents results for the full sample, which confirms the findings in Fig. 3. The trade-at rule increases the inverted venue share by 2.88% but decreases the dark venue share by 9.34%. Quoting at the nickel tick increases the inverted and dark share by 3.52% and 2.67%, respectively. All of those estimates are statistically significant. In contrast, the trade rule has no meaningful impact on market share. The economic significance of these changes can be observed by comparing the changes to the pre-pilot means reported in Table 1. The increase in inverted share due to the trade-at rule alone represents almost one-third of its pre-pilot period average level. The combined effect on inverted share due to the quote policy and the trade-at rule is 74% of the pre-pilot level. The negative impact of the trade-at rule on dark trading is partially offset by the increase associated with the quote rule. The net effect is a 17% decline in the level of dark trading. Coefficient estimates on other controls are consistent with our previous analysis of the determinants of market share.

We hypothesize that the observed shift in trading toward inverted venues is driven by liquidity providers seeking to gain queue priority by offering sub-tick price improvement (i.e., paying a maker fee on the inverted venue that is much smaller than a tick to gain execution priority). To see this, let *T* be the tick size, and assume that the midpoint is the underlying value of the asset. In the

¹² Our controls include control variables used in prior studies. O'Hara and Ye (2011) consider size as an important determinant in market fragmentation. Menkveld et al. (2017) use the VIX as an important determinant of lit and dark trading.

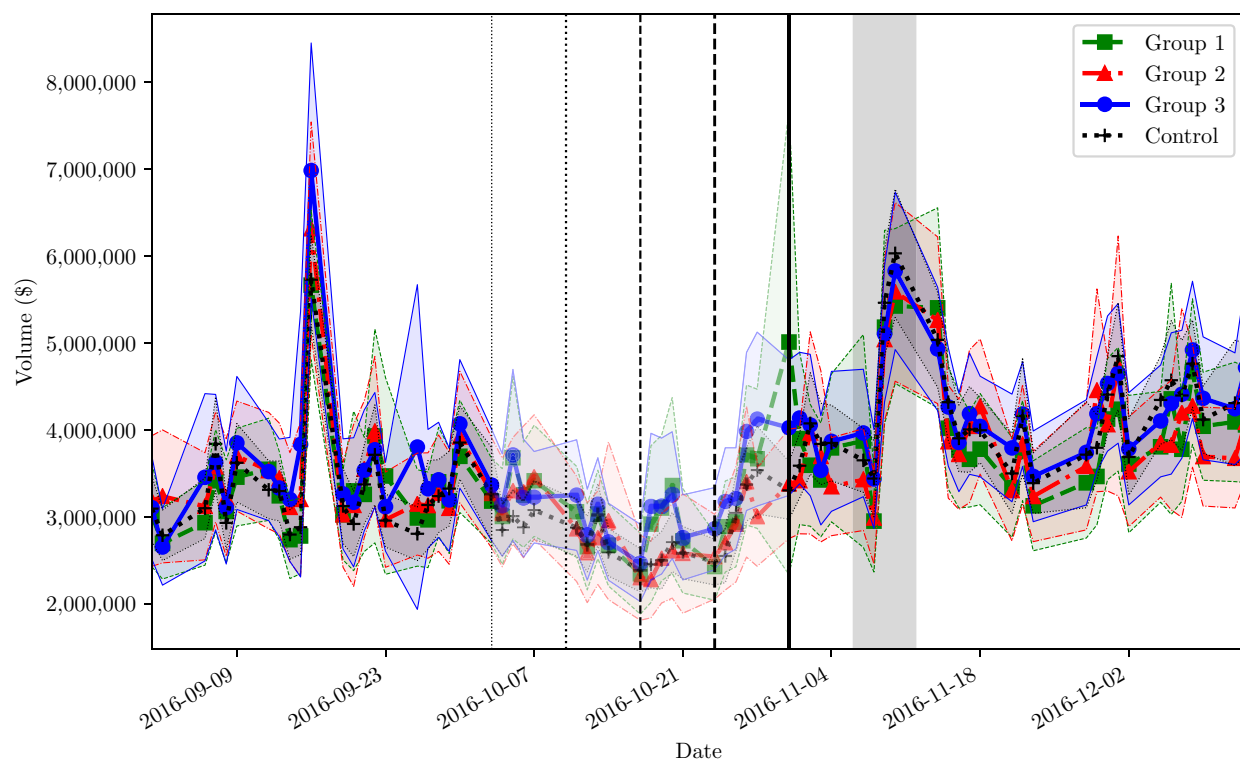


Fig. 2. Daily mean dollar volume. This figure shows the daily mean consolidated dollar volume from lit and dark venues. The means are presented separately for group one (green squares), group two (red triangles), group three (blue circles), and the control group (black crosses). The shaded areas are pointwise 95% confidence bands around the daily means. The vertical lines indicate when stocks are added to the pilot groups. Five stocks were added to groups one and two at the first line, a further 95 stocks were added at the second line, and all remaining group one and two stocks are added at the third line. Five group three stocks were also added at the third line, a further 95 group three stocks were added at the fourth line, and the remaining group three stocks were added at the fifth line. The dark shaded band indicates the US election week. The sample period is September 1 to December 13, 2016. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

absence of any fees, the effective spread received by liquidity providers (and paid by liquidity takers) for a one-way trade must lie on the grid $0.5T$, $1.5T$, $2.5T$, etc. For simplicity, consider both maker and taker fee equal to F (but with opposite signs). With an inverted fee structure, the grid of post-fee effective spreads becomes $0.5T - F$, $1.5T - F$, $2.5T - F$, etc., whereas with the traditional fee structure, the grid is $0.5T + F$, $1.5T + F$, $2.5T + F$, etc. The inverted fee structure allows the effective spread to be less than $0.5T$, while the traditional fee structure requires it always to be greater than $0.5T$.¹³ On inverted venues, liquidity providers receive a smaller effective spread by offering sub-tick price improvement. The distinction between inverted and traditional venues matters more when T is large enough that the ability to use an effective spread less than $0.5T$ matters to market participants. Hence, if our hypothesis is correct, the effect should be stronger for stocks that usually trade at a tighter spread. To validate this hypothesis, we split our sample stocks into low and high tick constraint groups based on whether their time-weighted average quoted spread during the prepilot period is above or

below the sample median (approximately 0.05), respectively.

Results from this subsample analysis paint a clearer picture of the dynamics at play. The impact of the trade-at rule on the inverted share is twice as large for high tick constraint stocks.¹⁴ This is consistent with our hypothesis that liquidity providers use inverted venues as a substitute to dark venues for sub-tick trading, which is more valuable among stocks that are tick constrained. We also find a stronger effect for the quote rule in high tick constraint stocks, although the relative difference is not as large as for the trade-at rule. The trade rule has no measurable impact.

5. Inverted venues and market quality

In the previous section, we find that market participants adapt to the tick size pilot by increasing the market share of inverted venues for pilot group stocks. The increased tick size increases inverted trading from around 9% to about 12%, and the implementation of the trade-at rule adds another 3%. These changes are consistent with our

¹³ As a special case, if F were equal to $0.5T$, the grids available with traditional and inverted fee structures would be identical except for the fact that the inverted fee structure allows an effective spread of zero.

¹⁴ Tests on differences between subsamples presented in Table A.2 of the Online Appendix confirm that this difference is statistically significant at the 1% level.

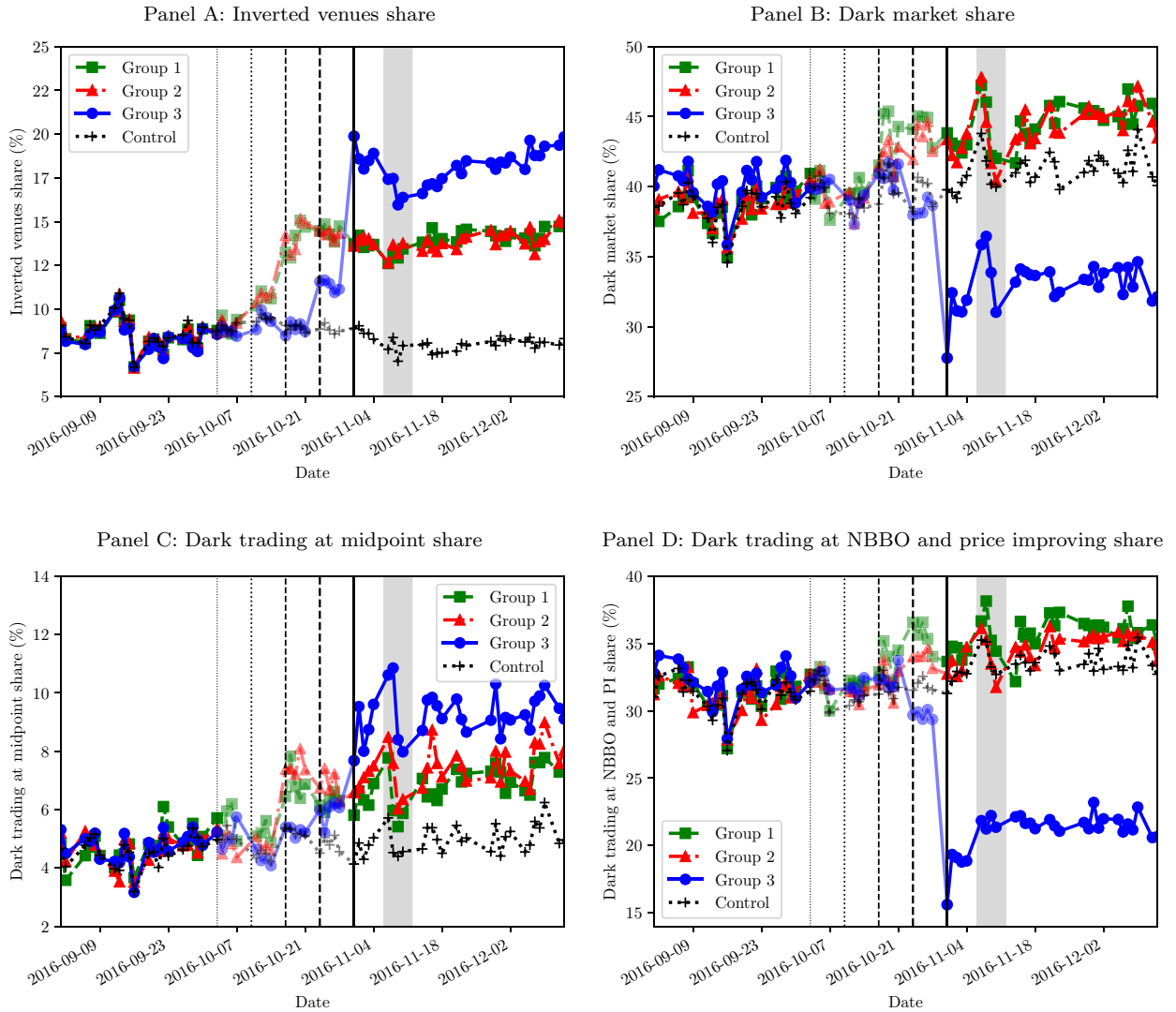


Fig. 3. Daily mean trading share. This figure shows the daily mean of the inverted venues market share (Panel A), the dark venues market share (Panel B), the market share of dark trading executed at the NBBO midpoint (Panel C), and the market share of non-midpoint dark trading that is at NBBO or price improving over the NBBO (Panel D). All values are in percentage terms, and market shares are computed relative to aggregate continuous trading. The means are presented separately for group one (green squares), group two (red triangles), group three (blue circles), and the control group (black crosses). The vertical lines indicate when stocks are added to the pilot groups. Five stocks were added to groups one and two at the first line, a further 95 stocks were added at the second line, and all remaining group one and two stocks are added at the third line. Five group three stocks are also added at the third line, a further 95 group three stocks were added at the fourth line, and the remaining group three stocks were added at the fifth line. The dark shaded band indicates the US election week. The sample period is September 1 to December 13, 2016. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

hypothesis that market participants increasingly use inverted venues to gain queue priority when the tick size increases and sub-tick trading is prohibited on dark venues. In this section, we study the impact of the increase in inverted trading share on market quality.

Specifically, we use the method of instrumental variables and run the following two-stage least squares (IV-2SLS) regression:

$$Y_{i,\tau} = \alpha + \beta_1 \widehat{\text{Inverted share}}_{i,\tau} + \beta_2 \widehat{\text{Dark share}}_{i,\tau} + \delta_1 \mathbf{1}_{[\text{trade}]_{i,\tau}} + \delta_2 \mathbf{1}_{[\text{quote}]_{i,\tau}} + \gamma X_{i,\tau} + \eta_{i,\tau}, \quad (2)$$

where $Y_{i,\tau}$ is a market quality measure. $\widehat{\text{Inverted share}}_{i,\tau}$ and $\widehat{\text{Dark share}}_{i,\tau}$ are the fitted market shares of inverted and dark venue trading from the first stage regressions, which use the specifications defined in Eqs. (1a) and (1b). We include dark share because results from the previous section show that it is also affected by the trade-at rule, and dark trading is known to affect market quality (e.g., Comerton-Forde and Putniņš, 2015; Foley and Putniņš, 2016). We measure market quality along two dimensions: pricing efficiency and liquidity. Table 4 reports the summary statistics for the efficiency and liquidity

Table 3

Market share. This table reports coefficients from OLS regressions of inverted venues and dark venues market share of dollar-volume continuous trading, both in %, on rule dummies ($1_{\text{trade-at}}$, 1_{trade} , and 1_{quote}), lags of the dependent variables, and controls. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), pilot group fixed effects, index membership fixed effects, industry fixed effects, and date fixed effects. Results are presented for the full sample and for two subsamples of stocks with below and above-median tick constraint. The level of tick constraint is measured as the inverse of the time-weighted average spread during the prepilot period of September 2016. *t*-statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, ** and, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. Sample descriptive statistics for the prepilot period are presented in Table 1.

	Full sample		Low tick constraint		High tick constraint	
	Inverted share (1)	Dark share (2)	Inverted share (3)	Dark share (4)	Inverted share (5)	Dark share (6)
$1_{\text{trade-at}}$	2.882*** (10.59)	−9.338*** (−23.57)	1.865*** (7.14)	−8.704*** (−14.82)	3.756*** (10.59)	−9.346*** (−20.27)
1_{trade}	−0.114 (−0.94)	−0.402 (−1.34)	−0.041 (−0.23)	−0.645 (−1.26)	−0.126 (−0.84)	−0.106 (−0.32)
1_{quote}	3.518*** (29.48)	2.671*** (9.78)	2.971*** (21.03)	1.448*** (3.13)	4.154*** (25.28)	3.465*** (12.97)
Inverted share ($t - 1$)	0.411*** (41.18)		0.350*** (31.80)		0.433*** (36.92)	
Dark share ($t - 1$)		0.245*** (38.76)		0.193*** (25.61)		0.333*** (39.47)
VIX	−0.156*** (−10.73)	1.143*** (29.55)	−0.188*** (−9.29)	1.237*** (21.86)	−0.030 (−1.53)	1.033*** (18.92)
$\text{Log}(\text{Size})$	1.014*** (20.24)	−2.317*** (−15.42)	1.102*** (14.14)	−2.500*** (−11.34)	0.616*** (9.47)	−2.149*** (−9.97)
$1/\text{Price}$	1.599*** (4.29)	10.064*** (5.67)	−0.013 (−0.02)	6.416*** (3.01)	−1.060** (−2.09)	9.149*** (3.74)
Pilot group F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Date F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.518	0.336	0.450	0.297	0.581	0.418
<i>N</i>	152,303	152,303	73,884	73,884	78,416	78,416

measures in the pre-pilot period. We discuss the construction of those measures in the following sections. In addition, we also include summary statistics for order activity metrics that are discussed in Section 6.

The tick size pilot provides an ideal setting to study how the interaction between the tick size, fee structures, and the trade-at rule affects market quality. The trade-at rule is a predetermined exogenous shock limiting sub-tick trading on dark venues but not on inverted venues. As a result, inverted venues gain popularity after the rule is implemented, in addition to the growth due to the tick size increase. Furthermore, the carefully designed experimental nature of the pilot guarantees that the effect of the trade-at rule on the pilot group is unrelated to the fundamental information environment of a stock. As a result, we can examine how changes in market participants' strategies, which takes the form of increased inverted trading, affect market quality. Our regression setting is an instrumental variables approach that uses the trade-at rule as an instrument to examine the impact of changes in inverted trading on market quality.

It is important to point out that the identification approach does not allow us to capture all aspects of inverted trading. We only study one potential channel through which inverted trading affects market quality, namely allowing traders to alter their queue priority for a fee. We

are unable to assess the impact of inverted venues on retail orders and order routing behavior of retail brokers because retail orders are partially exempt from the pilot rules, and we cannot observe broker routing decisions at the stock level. Although this limits our analysis on the overall impact of inverted trading, our test is sufficient to provide insights on how the sub-tick trading channel of inverted venues affects the market quality.

The instrumental variables approach allows us to address the problem of endogeneity in trading allocation regarding market conditions (see Buti et al., 2011 and Comerton-Forde and Putniņš, 2015 for detailed discussions on this issue). Since we have two endogenous variables, inverted and dark venue trading, we supplement the trade-at rule instrumental variable with the lagged inverted and dark market share. In our first stage regression, Eqs. (1a) and (1b), we include lagged inverted share or lagged dark share to instrument the market share of inverted and dark trading, respectively. This approach is similar to those adopted in Sarkar and Schwartz (2009) and Foley and Putniņš (2016). Given the experimental setup, it would be ideal to use one of the other pilot rule changes as a second instrument. However, we have shown that the trade rule has no explanatory power on inverted or dark trading share, and the quote rule is not a valid instrument because the change in the minimum tick size can mechanically

Table 4

Market measures statistics. This table presents the number of observations, mean, median, standard deviation, minimum and maximum values, and the 25th and 75th percentiles for variables of interest during the pre-pilot period of September 2016. All variables are daily stock-level observations. The sample includes all stocks from tick pilot groups and the control group. Price efficiency measures are the absolute difference of one with the variance ratio, the 15 s midpoint autocorrelation, and the 15 s market return cross-correlation. Liquidity measures are the time-weighted relative spread, the effective spread, the one-minute realized spread, and the one-minute price impact. The order activity measure is the trade-to-order ratio, aggregated for inverted and noninverted venues, using SEC MIDAS data. Impact of order activity variables are the 20-event cumulative impulse response functions associated with marketable and nonmarketable limit orders from a VAR estimated using ITCH data from NASDAQ and NASDAQ BX. The impact is presented per event and scaled by the daily number of events. The short selling measures are the dollar-volume short sale rates on NASDAQ BX and on dark venues and the share of short sales on NASDAQ BX for which the seller is the liquidity provider. See the text for details on variable construction.

	N	Mean	Std	Min	25 Pct	Median	75 Pct	Max
Price efficiency								
1 – Variance ratio (1 min/15 s)	47,711	0.16	0.15	0.00	0.05	0.12	0.23	1.39
Autocorrelation (15 s)	47,711	0.07	0.07	0.00	0.02	0.05	0.10	0.85
Cross-correlation (15 s)	47,693	0.07	0.05	0.00	0.02	0.06	0.11	0.31
Liquidity								
Time-weighted relative spread (%)	47,711	0.84	1.38	0.06	0.18	0.35	0.85	13.07
Effective spread (all venues, %)	47,471	0.53	0.88	0.04	0.11	0.22	0.54	7.61
Realized spread, 1 min (all venues, %)	47,471	0.29	0.79	–1.09	–0.00	0.05	0.25	6.90
Price impact, 1 min (all venues, %)	47,471	0.24	0.38	–0.41	0.08	0.13	0.24	3.47
Order activity								
Trade volume to order volume (inverted venues, %)	27,874	3.60	3.71	0.04	1.48	2.53	4.40	74.63
Trade volume to order volume (noninverted venues, %)	36,169	2.80	2.28	0.02	1.27	2.32	3.68	47.40
Impact of order activity								
Market. limit order (NASDAQ)	29,278	0.27	0.30	–0.06	0.09	0.17	0.32	2.80
Market. limit order (NASDAQ BX)	29,278	0.09	0.13	–0.16	0.03	0.05	0.11	1.66
Nonmarket. limit order at NBBO (NASDAQ)	29,278	0.32	0.28	–0.19	0.15	0.24	0.41	2.02
Nonmarket. limit order at NBBO (NASDAQ BX)	29,278	0.20	0.19	–0.15	0.08	0.14	0.25	1.77
Market. limit order $\times$ N events (NASDAQ)	29,278	78.73	172.77	–18.88	22.77	44.21	85.79	15,584.80
Market. limit order $\times$ N events (NASDAQ BX)	29,278	4.56	10.07	–20.37	0.73	1.88	4.73	661.57
Nonmarket. limit order at NBBO $\times$ N events (NASDAQ)	29,278	954.85	841.30	–857.56	439.71	737.34	1,206.02	25,387.22
Nonmarket. limit order at NBBO $\times$ N events (NASDAQ BX)	29,278	159.77	200.46	–53.03	43.28	101.53	201.50	6,136.58
Short selling								
Short rate on NASDAQ BX (%)	46,319	1.07	1.31	0.00	0.29	0.88	1.49	100.00
Short rate on dark venues (%)	46,319	15.96	10.56	0.00	8.85	13.77	20.45	291.96
Share of short sales by liquidity providers (%)	35,363	55.47	26.48	0.00	38.89	55.56	73.33	100.00

affect our market quality measures. Therefore, we cannot use these rule changes as instruments.

The use of the instrumental variables approach often raises concerns as to the validity of the instruments or appropriateness of the approach (see, e.g., Roberts and Whited, 2013; Jiang, 2017). We discuss potential issues with the approach used for our main results in Section 4 of the Online Appendix. Specifically, we repeat our main tests using uninstrumented OLS regressions and discuss the exclusion restriction and the use of lagged variables. Overall, our robustness tests suggest that these potential issues are not problematic in our setup.

5.1. Pricing efficiency

We begin our analysis of market quality with pricing efficiency, using a set of high-frequency proxies. These include the variance ratio, the absolute value of the return autocorrelation, and the absolute value of the cross-correlation between stock and market returns. Following O'Hara and Ye (2011), we use the absolute difference between one and the variance ratio to proxy for information efficiency:

$$|1 - VR| = \left| 1 - \frac{\sigma_{1\text{min};i,\tau}^2}{4\sigma_{15\text{sec};i,\tau}^2} \right|,$$

where $\sigma_{15\text{sec};i,\tau}^2$ and $\sigma_{1\text{min};i,\tau}^2$ are daily variances of NBBO midpoint returns for stock i on day τ sampled at 15 s and one minute intervals, respectively.¹⁵ The second high-frequency measure of pricing efficiency is the absolute value of NBBO midpoint return autocorrelation (AC). We choose a 15 s interval to estimate the autocorrelation to maintain consistency with the interval choice in the variance ratio estimation. The intuition of these two measures is that if the midpoint is the market estimate of the value of the stock, an efficient price process implies that the midpoint follows a random walk, with variance ratio equal to one and autocorrelation equal to zero. The last efficiency measure is the absolute value of the cross-correlation between the stock return and the lagged return of its corresponding index (CC). Specifically, we estimate the correlation between the stock NBBO midpoint return and the lagged NBBO midpoint return of the iShares Russell 2000 ETF. The time interval for estimation is 15 s. This efficiency measure captures the delay in the adjustment to market-wide information. This delay is minimized in an efficient market. For all three measures mentioned above, a smaller value corresponds to better pricing efficiency.

¹⁵ We also consider variance ratios at different horizons up to 30 min over 5 min. Results, presented in Table A.4 of the Online Appendix, are robust to the choice of the horizon.

Table 5

Pricing efficiency. This table reports coefficients from IV-2SLS regressions of price efficiency measures on inverted venues market share and dark venues market share, both in decimals, rule dummies ($\mathbf{1}_{\text{trade}}$, and $\mathbf{1}_{\text{quote}}$), and controls. The price efficiency measures are the absolute difference between one and the variance ratio ($|1 - \text{VR}|$), the 15 s midpoint return autocorrelation (AC (15 s)), and the 15 s midpoint return market cross-correlation (CC (15 s)). Instrumented variables inverted venues market share and dark venues market share are estimated in the first stage by regressing on the dependent variables of the second stage, using the trade-at rule dummy ($\mathbf{1}_{\text{trade-at}}$) and one lag of the instrumented variable as instruments. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), pilot group fixed effects, index membership fixed effects, industry fixed effects, and date fixed effects. Results are presented for the full sample and for two subsamples of stocks with below and above-median tick constraint. The level of tick constraint is measured as the inverse of the time-weighted average spread during the pre-pilot period of September 2016. t -statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, ** and, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. See the text for details on variable construction. Sample descriptive statistics for the pre-pilot period are presented in Tables 1 and 4.

	Full sample			Low tick constraint			High tick constraint		
	$ 1 - \text{VR} $ (1)	AC (15 s) (2)	CC (15 s) (3)	$ 1 - \text{VR} $ (4)	AC (15 s) (5)	CC (15 s) (6)	$ 1 - \text{VR} $ (7)	AC (15 s) (8)	CC (15 s) (9)
Inverted share	−0.099*** (−3.34)	−0.035** (−2.46)	−0.037*** (−3.18)	−0.024 (−0.49)	−0.019 (−0.77)	0.014 (1.03)	−0.167*** (−4.28)	−0.056*** (−3.30)	−0.083*** (−4.85)
Dark share	−0.108*** (−6.71)	−0.061*** (−7.12)	−0.024*** (−4.70)	−0.122*** (−5.05)	−0.072*** (−5.18)	−0.004 (−0.83)	−0.077*** (−4.05)	−0.045*** (−5.11)	−0.057*** (−7.34)
$\mathbf{1}_{\text{trade}}$	0.003 (0.73)	0.002 (1.17)	−0.002 (−1.43)	0.004 (0.68)	0.003 (0.83)	−0.001 (−0.65)	0.005 (1.03)	0.003 (1.28)	−0.004* (−1.80)
$\mathbf{1}_{\text{quote}}$	−0.001 (−0.18)	−0.006*** (−3.26)	−0.009*** (−5.72)	−0.017*** (−3.42)	−0.013*** (−4.60)	−0.002 (−1.37)	0.016*** (3.34)	0.001 (0.36)	−0.015*** (−5.67)
VIX	0.004*** (13.63)	0.002*** (12.46)	−0.002*** (−7.15)	0.003*** (7.86)	0.002*** (6.16)	−0.003*** (−9.84)	0.004*** (8.12)	0.002*** (9.63)	−0.000 (−0.74)
$\text{Log}(\text{Size})$	−0.008*** (−6.81)	−0.004*** (−5.74)	0.016*** (17.48)	−0.003 (−1.41)	−0.001 (−1.09)	0.014*** (13.29)	−0.008*** (−5.97)	−0.005*** (−6.20)	0.016*** (14.33)
$1/\text{Price}$	−0.014 (−1.61)	−0.015** (−2.59)	0.013*** (3.32)	0.062*** (3.21)	0.005 (0.49)	0.039*** (5.79)	−0.016* (−1.70)	−0.013** (−2.15)	−0.010* (−1.73)
Pilot group F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.011	0.016	0.371	0.011	0.017	0.354	0.013	0.016	0.398
N	152,303	152,303	152,172	73,884	73,884	73,800	78,416	78,416	78,369

Table 5 presents coefficient estimates of our IV-2SLS regressions of pricing efficiency. t -statistics from robust standard errors clustered by date and by stock are presented in parenthesis. Coefficient estimates on the market share of inverted venues are negative and significant, indicating that trading on inverted venues increases pricing efficiency. The improvement in efficiency is economically meaningful, as a 10% increase of the inverted share reduces $|1 - \text{VR}|$ and AC by 0.0099 and 0.0035, respectively. This corresponds to reductions of about 6.2% and 5.0% of each measure's pre-pilot levels. This improvement in pricing efficiency due to trading on inverted venues is consistent with our hypothesis that liquidity providers use these venues to gain queue priority for a fee. However, if this channel is at play, we expect the impact to be concentrated in stocks for which the tick size is binding. To test this, we repeat our tests for low and high tick constraint stocks separately. We confirm that the beneficial impact on pricing efficiency is only present in highly tick constrained stocks.¹⁶

Coefficient estimates on the dark share are also negative and significant, suggesting that an increase in dark trading is beneficial for pricing efficiency. This result is consistent with Buti et al. (2011) and Foley and Putniņš (2016), who find that dark trading can improve market efficiency. This result holds for both high and low tick constraint stocks, except for the coefficient estimate of CC on low tick constraint stocks. Finally, consistent with prior literature, controls also confirm that larger stocks offer better pricing efficiency.

We find that the impact of the quote rule is mixed and varies between subsamples. We do not find that the trade rule has any significant impact on pricing efficiency. This is universal across our subsamples and over different efficiency measures. We discuss the impact of the different pilot rules on market quality in more details at the end of this section.

5.2. Liquidity

We next shift our focus to liquidity using four daily proxies: relative spread, effective spread, realized spread, and price impact. Relative spread is defined as the time-weighted spread divided by the prevailing midpoint. Effective spread, realized spread, and price impact of individual

¹⁶ Tests on differences between subsamples presented in Table A.3 of the Online Appendix confirm that this difference is statistically significant at the 5% level for the variance ratio and at the 1% level for the cross-correlation.

Table 6

Liquidity. This table reports coefficients from IV-2SLS regressions of liquidity measures on inverted venues market share and dark venues market share, both in decimals, rule dummies (1_{trade} , and 1_{quote}), and controls. The liquidity measures are the relative spread, the effective spread, the one-minute realized spread, and the one-minute price impact. Instrumented variables inverted venues market share and dark venues market share are estimated in the first stage by regressing on the dependent variables of the second stage, using the trade-at rule dummy ($1_{\text{trade-at}}$) and one lag of the instrumented variable as instruments. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), pilot group fixed effects, index membership fixed effects, industry fixed effects, and date fixed effects. Results are presented for the full sample and for two subsamples of stocks with below and above-median tick constraint. The level of tick constraint is measured as the inverse of the time-weighted average spread during the pre-pilot period of September 2016. *t*-statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. See the text for details on variable construction. Sample descriptive statistics for the pre-pilot period are presented in Tables 1 and 4.

	Full sample				Low tick constraint				High tick constraint			
	Rel. spd. (1)	Eff. spd. (2)	Real. spd. (3)	Prc. impact (4)	Rel. spd. (5)	Eff. spd. (6)	Real. spd. (7)	Prc. impact (8)	Rel. spd. (9)	Eff. spd. (10)	Real. spd. (11)	Prc. impact (12)
Inverted share	−5.687*** (−10.33)	−3.066*** (−11.16)	−2.438*** (−11.44)	−0.643*** (−6.39)	−6.101*** (−6.64)	−3.332*** (−7.11)	−2.693*** (−7.38)	−0.697*** (−3.99)	−2.709*** (−7.68)	−1.743*** (−9.17)	−1.325*** (−9.03)	−0.394*** (−4.23)
Dark share	−0.698*** (−2.77)	−0.293** (−2.19)	0.020 (0.17)	−0.360*** (−6.09)	0.070 (0.18)	0.064 (0.30)	0.361* (1.93)	−0.412*** (−4.08)	−0.760*** (−4.78)	−0.354** (−3.52)	−0.076 (−1.01)	−0.253*** (−4.61)
1_{trade}	0.096** (2.41)	0.048* (1.93)	0.040** (2.10)	0.002 (0.17)	0.118* (1.87)	0.080** (2.33)	0.068** (2.28)	0.002 (0.11)	0.043 (1.10)	0.003 (0.12)	0.001 (0.07)	0.002 (0.19)
1_{quote}	0.558*** (11.07)	0.362*** (13.06)	0.254*** (11.33)	0.113*** (10.28)	0.268*** (3.82)	0.176*** (4.87)	0.145*** (4.71)	0.039** (2.63)	0.630*** (13.81)	0.455*** (14.91)	0.290*** (12.77)	0.163*** (12.45)
VIX	0.174*** (21.61)	0.096*** (20.25)	0.057*** (15.04)	0.038*** (22.76)	0.178*** (14.18)	0.096*** (12.85)	0.056*** (8.91)	0.040*** (14.23)	0.066*** (8.59)	0.039*** (7.06)	0.016*** (4.60)	0.022*** (10.15)
$\text{Log}(\text{Size})$	−0.523*** (−19.33)	−0.301*** (−19.48)	−0.189*** (−16.18)	−0.107*** (−20.24)	−0.530*** (−12.29)	−0.301*** (−12.43)	−0.189*** (−10.07)	−0.108*** (−12.57)	−0.188*** (−7.08)	−0.118*** (−5.95)	−0.055*** (−4.31)	−0.062*** (−8.14)
$1/\text{Price}$	−0.492** (−2.22)	0.273* (1.81)	−0.031 (−0.28)	0.320*** (5.57)	1.483** (2.42)	1.557*** (4.18)	0.873*** (3.01)	0.660*** (6.43)	1.474*** (3.57)	1.270*** (4.03)	0.737*** (3.69)	0.537*** (4.47)
Pilot group F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.337	0.371	0.222	0.184	0.324	0.356	0.213	0.154	0.491	0.572	0.323	0.296
N	152,303	151,961	151,961	151,961	73,884	73,558	73,558	73,558	78,416	78,400	78,400	78,400

trades are estimated as follows:

$$\text{Effective spread} = 2 \cdot |p_{\tau} - m_{\tau}|, \quad (3)$$

Realized spread

$$= \begin{cases} 2(p_{\tau} - m_{\tau+1\min}) & \text{for buyer-initiated trades,} \\ 2(m_{\tau+1\min} - p_{\tau}) & \text{for seller-initiated trades,} \end{cases} \quad (4)$$

Price impact

$$= \begin{cases} 2(m_{\tau+1\min} - m_{\tau}) & \text{for buyer-initiated trades,} \\ 2(m_{\tau} - m_{\tau+1\min}) & \text{for seller-initiated trades,} \end{cases} \quad (5)$$

where p_t is the log traded price for the transaction at time t , m_t is the prevailing log midpoint at time t , and τ is the trade time. Trades are signed according to the Lee and Ready (1991) algorithm, using the prevailing quote one millisecond prior to the trade, and trades at the midpoint are signed using the tick test. The daily measures are the value-weighted averages for all trades occurring during continuous trading hours. All four liquidity proxies are high-frequency measures capturing the cost of trading. Relative spread captures a round-trip cost at the top of the limit order book, and effective spread is the cost of immediacy for marketable orders. Bessembinder and Venkatara-

man (2010) show the total price impact can be decomposed into a temporary impact captured by the realized spread and a permanent impact captured by the price impact measure. For all measures, a lower value corresponds to better liquidity.

We repeat the exercise from the previous section, using the IV-2SLS specification described in Eq. (2) to assess the effect of inverted and dark trading on liquidity. Estimation results presented in Table 6 indicate that increases in inverted share improve liquidity as coefficients are negative and statistically significant. This is universal for the four liquidity measures, and the effect is economically significant. For example, a one percent increase in the inverted share reduces the relative spread by 5.7 basis points (bps) (or 6.8% of its pre-pilot mean), the effective spread by 3.1 bps (5.8% of its pre-pilot mean), the realized spread by 2.4 bps (8.3% of its pre-pilot mean), and the price impact by 0.6 bps (2.7% of its pre-pilot mean). This finding is consistent in the analysis of subsamples split on the level of tick constraint. Although we find that dark share has a similar impact on liquidity, the effect is economically smaller and less significant.

Our results on pricing efficiency and liquidity suggest that increased trading on inverted venues induced by the trade-at rule has overall beneficial effects on market quality, particularly in tick constrained stocks. We

hypothesize that this beneficial effect arises from changes in routing strategies. Inverted venues attract liquidity demanders with rebates and enable liquidity providers to gain queue priority by paying a fee. Since the fee is smaller than a tick, this is effectively a sub-tick price improvement strategy for liquidity providers. When dark venues are prohibited from offering sub-tick price improvement, inverted venues become the only intermediaries able to fulfill traders' demand for queue priority through sub-tick price improvement. Information from those traders, which would otherwise be lost, is retained. This is even more pronounced in tick constrained stocks, for which sub-tick price improvement is the only way liquidity providers can compete, other than at the midpoint. With more information released to the market, pricing efficiency and market liquidity are improved.

5.3. Net policy effects

Our results so far show that the tick size pilot affects the market share of both inverted and dark trading and that both these market shares affect market quality. In this section, we study the net effect on market quality of the pilot policy changes. We discuss these effects in the context of our research question and refer the reader to [Rindi and Werner \(2017\)](#) for an in-depth analysis of the impact of the tick size pilot. Most importantly, we compare the net policy effect with the effect of trade-at induced inverted trading on market quality, which highlights not only the role of inverted trading in isolation but also the net effects of shifts in trading from dark venues to inverted venues. In addition to the previously introduced measures of pricing efficiency and liquidity, we also present results for fragmentation, measured as the inverse Herfindahl index of lit trading venues during continuous trading.

To assess the impact of the different trading rules on market quality, we regress pricing efficiency and liquidity measures on redefined rule dummies:

$$Y_{i,\tau} = \alpha + \beta_1 \mathbf{1}_{[\text{s.t. trade-at}]_i} + \beta_2 \mathbf{1}_{[\text{s.t. trade}]_i} + \beta_3 \mathbf{1}_{[\text{s.t. quote}]_i} + \beta_4 \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_5 \mathbf{1}_{[\text{s.t. trade-at}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_6 \mathbf{1}_{[\text{s.t. trade}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_7 \mathbf{1}_{[\text{s.t. quote}]_{i,\tau}} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \gamma X_{i,\tau} + \epsilon_{i,\tau}, \quad (6)$$

where $Y_{i,\tau}$ is a market quality measure. $\mathbf{1}_{[\text{s.t. trade-at}]_i}$, $\mathbf{1}_{[\text{s.t. trade}]_i}$ and $\mathbf{1}_{[\text{s.t. quote}]_i}$ are dummy variables equal to one if stock i is subject to the rule at any time during the sample, and zero otherwise. $\mathbf{1}_{[\text{post}]_{i,\tau}}$ is a dummy variable equal to one if the pilot rule for stock i is activated at time τ , and zero otherwise. For control stocks, the activation date is randomly set to one of the five possible date with probabilities matching the empirical distribution for pilot stocks.¹⁷

Our regression model is effectively a difference-in-difference analysis with rule dummies capturing the impact of pilot rules on market quality. We also run a triple-diff regression to compare the impact of pilot policies

between low and high tick constraint stocks:

$$Y_{i,\tau} = \alpha + \beta_1 \mathbf{1}_{[\text{s.t. trade-at}]_i} + \beta_2 \mathbf{1}_{[\text{s.t. trade}]_i} + \beta_3 \mathbf{1}_{[\text{s.t. quote}]_i} + \beta_4 \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_5 \mathbf{1}_{[\text{tick const.}]_i} + \beta_6 \mathbf{1}_{[\text{post}]_{i,\tau}} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_7 \mathbf{1}_{[\text{s.t. trade-at}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_8 \mathbf{1}_{[\text{s.t. trade}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_9 \mathbf{1}_{[\text{s.t. quote}]_{i,\tau}} \times \mathbf{1}_{[\text{post}]_{i,\tau}} + \beta_{10} \mathbf{1}_{[\text{s.t. trade-at}]_i} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_{11} \mathbf{1}_{[\text{s.t. trade}]_i} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_{12} \mathbf{1}_{[\text{s.t. quote}]_i} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_{13} \mathbf{1}_{[\text{s.t. trade-at}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_{14} \mathbf{1}_{[\text{s.t. trade}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} \times \mathbf{1}_{[\text{tick const.}]_i} + \beta_{15} \mathbf{1}_{[\text{s.t. quote}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} \times \mathbf{1}_{[\text{tick const.}]_i} + \gamma X_{i,\tau} + \epsilon_{i,\tau}. \quad (7)$$

We capture the comparison with the interaction term $\mathbf{1}_{[\text{s.t. rule}]_i} \times \mathbf{1}_{[\text{post}]_{i,\tau}} \times \mathbf{1}_{[\text{tick const.}]_i}$, where $\mathbf{1}_{[\text{tick const.}]_i}$ is a dummy variable equal to one if stock i is in the high tick constraint group, and zero otherwise. As before, stocks are deemed to be tick constrained if they have below-median time-weighted average spread during the pre-pilot period.

We present estimates of pricing efficiency and liquidity measures on rules dummies in [Table 7](#). The net effect of the trade-at rule appears to be negative as pricing efficiency decreases according to the variance ratio and the 15 s autocorrelation, and liquidity deteriorates according to the price impact, while changes are not statistically significant for other measures.

An interesting result emerges when we compare the net policy effect of the trade-at rule with the effect of the trade-at induced inverted trading. While the increased inverted trading due to the trade-at rule is beneficial, the trade-at rule is detrimental to market quality. The key to reconcile these findings is noticing that the dark market share drops following the trade-at rule. We know from [Table 3](#) that the inverted venue and dark venue market share are both affected by the increase in the minimum tick size and the trade-at rule, albeit differently.¹⁸ The increase in the tick size increases both inverted and dark share, while the imposition of the trade-at rule increases inverted trading but decreases dark trading, and the latter is larger than the former (2.6% versus −8.6%). Furthermore, past studies have shown that dark trading can be beneficial to pricing efficiency. This suggests that the improvement in pricing efficiency due to the increase in inverted share is not enough to compensate for the decrease due to the lower dark share.

Another difference arises when we compare the effect on low versus high tick constraint stocks. The trade-at rule impacts both low and high tick constraint stocks equally. But when we turn to the effect of inverted trading, the increase in inverted trading in high tick constraint stocks improves market quality more than it does in low tick constraint stocks because there is more demand for sub-tick trading in high tick constraint stocks. The benefits to market quality of trading on inverted venues are thus likely the result of a substitution effect, where inverted venues

¹⁷ See [Section 5](#) of the Online Appendix for more details on the random assignment procedure.

¹⁸ We also repeat the analysis presented in [Table 3](#) using the difference-in-difference approach. Results, presented in [Section I.e](#) of the Online Appendix, are qualitatively similar to those presented in [Table 3](#).

Net policy effects. This table reports coefficients from OLS regressions of market quality measures. The price efficiency measures are the absolute difference between one and the variance ratio ($(1 - VR)$), the 15 s midpoint return autocorrelation (AC (15 s)), and the 15 s midpoint return market cross-correlation (CC (15 s)). The liquidity measures are the relative spread, the effective spread, the one-minute realized spread, and the one-minute price impact. $\mathbf{1}_{\{s.t. trade-at\}}$, $\mathbf{1}_{\{s.t. trade\}}$, $\mathbf{1}_{\{s.t. quote\}}$ are dummy variables equal to one if the stock is subject to the rule at one point in the sample, and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. $\mathbf{1}_{\{tick const\}}$ is a dummy variable equal to one if the stock has above-median tick constraint, and zero otherwise. The level of tick constraint is measured as the inverse of the time-weighted average spread during the prepilot period of September 2016. $\mathbf{1}_{\{post\}}$ is a dummy variable equal to one if the stock is in the posttreatment period, and zero otherwise. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), index membership fixed effects, and industry fixed effects. t -statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016 to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. See the text for details on variable construction. Sample descriptive statistics for the prepilot period are presented in [Tables 1](#) and [4](#).

[illegible]

replace dark venues as the preferred platform for liquidity competition using sub-tick trading.

As for the quote rule, which increases the tick size, the net result is better pricing efficiency along all dimensions. On the flip side, increasing the minimum tick size is harmful to liquidity, which is consistent with the findings of Yao and Ye (2018). That said, the increase in the realized spread following the quote rule suggests that the profit margin for market making increases. This is consistent with the results of Rindi and Werner (2017) who also examine market makers' profits around the tick size pilot. This aligns with one of the intended outcomes of the pilot, i.e., greater profits for liquidity provision by market makers, who may, in turn, provide better trading and research support for small companies. All these effects are statistically significant. Consistent with Chao et al. (2017), we find that fragmentation increases significantly. Finally, we do not find any statistically significant impact on our market quality measures resulting from the imposition of the trade rule.

Our net policy results are consistent with other papers that study the impact of tick size pilot on market quality (e.g., Rindi and Werner, 2017; Albuquerque et al., 2017; Lin et al., 2017). The important contribution of our paper is to highlight the increase in inverted trading resulting from the pilot and study its impact on market quality. This helps us understand the economics of the interaction between the tick size, fee structures, and the trading rule, which are important features of the current market landscape.

6. Liquidity competition on inverted venues

In this section, we zoom in on order activity to test the hypothesis that inverted trading induced by the trade-at rule improves market quality by enabling competition between traders, in particular, informed liquidity providers.

When traders place an order, they choose the order type—a marketable or nonmarketable limit order—and where to route the order. In the order placement decision, traders make trade-offs between the execution price and execution probability. The pricing grid is an essential determinant of this trade-off. In a fragmented market with various fee structures, the pricing grid of a venue is jointly determined by its fee structure and the tick size. Therefore, the fee structure and the tick size jointly affect the choice of order type and the routing decision. For example, a larger tick size makes crossing the spread with a marketable order more costly and makes a nonmarketable limit order more attractive. The resulting increase in competition for liquidity provision increases the queue length, which reduces the probability of execution and the benefits of nonmarketable limit orders.

Different fee structures add another layer to the order placement decision. A larger tick size and a longer queue will encourage traders to use inverted and dark venues where liquidity providers can submit orders within the spread and join a separate queue. This separate queue has a higher probability of execution when the price improvement for liquidity takers is attractive enough, leading to a pecking order in the spirit of Menkveld et al. (2017). The decision to route orders to inverted and dark venues re-

sults in the same trade-off between price and execution probability. This trade-off is more important for tick constrained stocks because the cost of crossing the spread with a marketable order is higher and the probability of execution for a nonmarketable order is lower.

To highlight the impact of inverted fee structure on order placement and liquidity provision, we take advantage of the incremental change in pilot rules. More specifically, both the quote rule and the trade rule contribute to increasing the tick size and also make inverted venues more attractive. However, the trade-at rule does not affect the tick size but increases the attractiveness of inverted venues, as liquidity providers are more constrained from dark trading. By focusing on the impact the trade-at rule has on inverted venue order activity, we capture the effect of the fee structure on order placement strategies.

For the rest of the section, we first look at the contribution to displayed liquidity at NBBO by nonmarketable limit orders posted on inverted venues and the fraction of executions on these venues. We then look at changes to the trade-to-order ratio, a proxy for execution probability, for both inverted and noninverted venues around the pilot. Finally, we look at the information contribution of orders sent to inverted venues using a vector autoregression (VAR) approach and the activity of informed traders as proxied by short selling.

6.1. Liquidity supply and market share

We first study liquidity supply and the fraction of executions on inverted venues based on individual exchange-specific data obtained from TRTH. Fig. 4 shows how the liquidity supply varies over time on inverted venues. Panels A to C present the daily average relative depth contribution of BATS-Y, NASDAQ BX, and EDGA, respectively. We define the relative depth as the time-weighted average NBBO depth available on an individual venue, divided by the total NBBO depth available across all lit venues. We find that the average relative depth decreases for BATS-Y and EDGA, while it increases for NASDAQ BX for all pilot groups relative to the control group. Nonetheless, these changes are small in magnitude. Panels D to E present the daily average fraction of the time that each exchange is quoting at the NBBO. This measure increases in a similar way for all pilot groups, while it remains constant for the control group. BATS-Y and NASDAQ BX go from quoting at NBBO around 20% of the time to above 50% of the time, while EDGA goes from 10% to over 30%. This finding highlights that liquidity providers increasingly quote on inverted venues. They are willing to offer an implicit price improvement with the inverted fee structure in return for a higher execution probability.

Fig. 5 shows the fraction of executions on inverted venues, in percentage terms. Panels A to C show the fraction of executions on inverted venues unconditionally, conditional on having at least one inverted venue at the NBBO, and conditional on having all inverted venues at the NBBO, respectively. All fractions are estimated daily at the group level for each pilot group and the control group. For example, in Panel B, the fraction for group three is computed by first selecting all trades of group three stocks executed

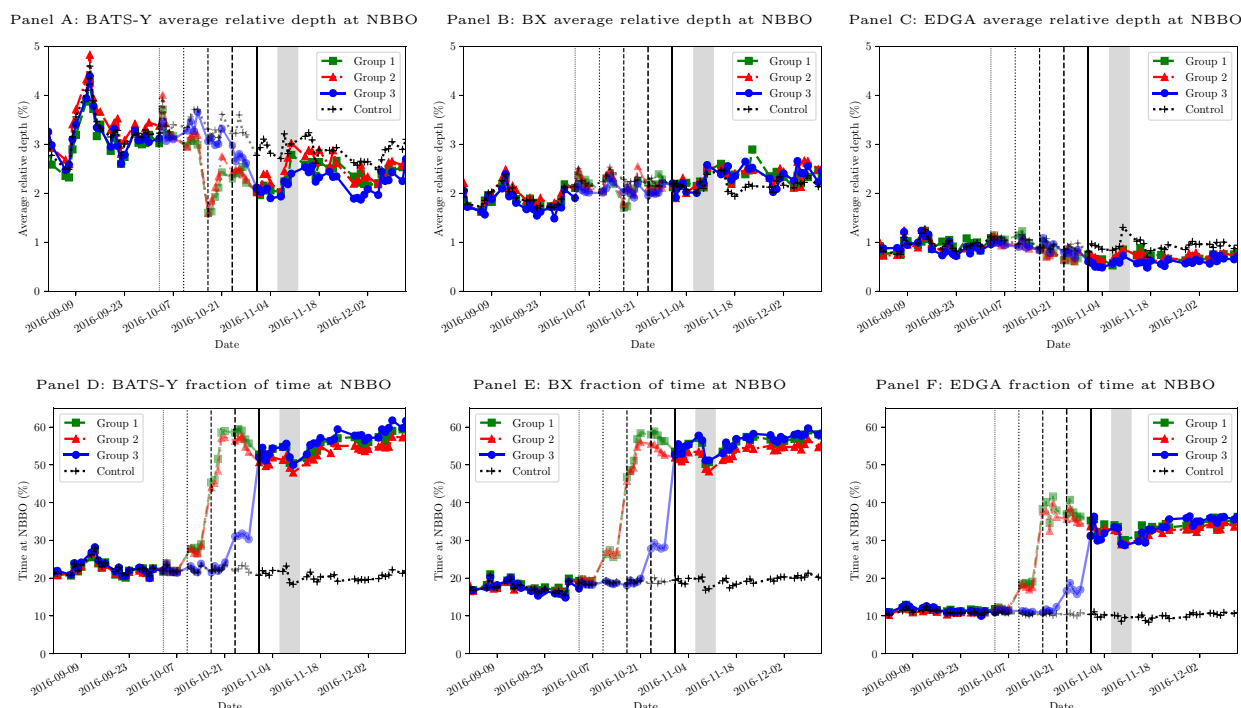


Fig. 4. Inverted venues quotes at NBBO. This figure shows the daily mean relative depth at NBBO for BATS-Y, NASDAQ BX, and EDGA (Panels A, B, and C, respectively) and the daily mean fraction of time at NBBO for the same venues (Panels D, E, and F, respectively). The relative depth is defined as the time-weighted average quantity at NBBO on inverted venues, divided by the total quantity at NBBO on all lit venues. All values are in percentage terms. The means are presented separately for group one (green squares), group two (red triangles), group three (blue circles), and the control group (black crosses). The vertical lines indicate when stocks are added to the pilot groups. Five stocks were added to groups one and two at the first line, a further 95 stocks were added at the second line, and all remaining group one and two stocks are added at the third line. Five group three stocks are also added at the third line, a further 95 group three stocks were added at the fourth line, and the remaining group three stocks were added at the fifth line. The dark shaded band indicates the US election week. The sample period is September 1 to December 13, 2016. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

when at least one inverted venue is quoting at NBBO and then computing the fraction of those trades that are executed on an inverted venue. We find that inverted venues attract liquidity takers as they receive over 45% of trades when at least one of them is at NBBO and over 50% of trades when they all are at NBBO. These fractions increase for all pilot groups after the pilot is enacted and much more so for group three that is subject to the trade-at rule. This suggests not only that liquidity providers are more active on inverted venues, but that liquidity demanders also respond by routing more orders to these venues.

We investigate further, looking at the fraction of executions on individual venues. Panels D to F decompose the contribution of each inverted venue to the conditional fractions in Panel C. This decomposition presents two notable patterns. First, the pilot increases the fraction of executions on BATS-Y and NASDAQ BX for all groups but decreases it slightly on EDGA. This effect is consistent with Fig. 4, which shows more liquidity provided on BATS-Y and NASDAQ BX after the pilot implementation. This is also expected, as BATS-Y and NASDAQ BX offer larger rebates and should thus attract more rebate-seeking marketable orders. Second, the increase in the fraction is more pronounced for NASDAQ BX than BATS-Y, especially for group three.

After the trade-at rule is implemented, both relative depth and fractions of executions on inverted venues increase, which indicates that more marketable and non-marketable limit orders are routed to inverted venues. The pattern is consistent with our hypothesis that liquidity providers use the inverted fee structure to offer an implicit price improvement (the maker fee) in exchange for queue priority.

6.2. Trade-to-order

We next examine the trade-to-order ratio for individual exchanges based on data from SEC MIDAS.¹⁹ Though some of the past studies use the trade-to-order ratio as a proxy for algorithmic or high-frequency trading (HFT) activity (e.g., Hendershott et al., 2011, Buti et al., 2015), we do not use it to infer trader type activity nor market

¹⁹ MIDAS, which stands for Market Information Data Analytics System, is a data set provided by the SEC summarizing order activity in equity trading on a daily basis. Details are available at <https://www.sec.gov/marketstructure/midas.html>.

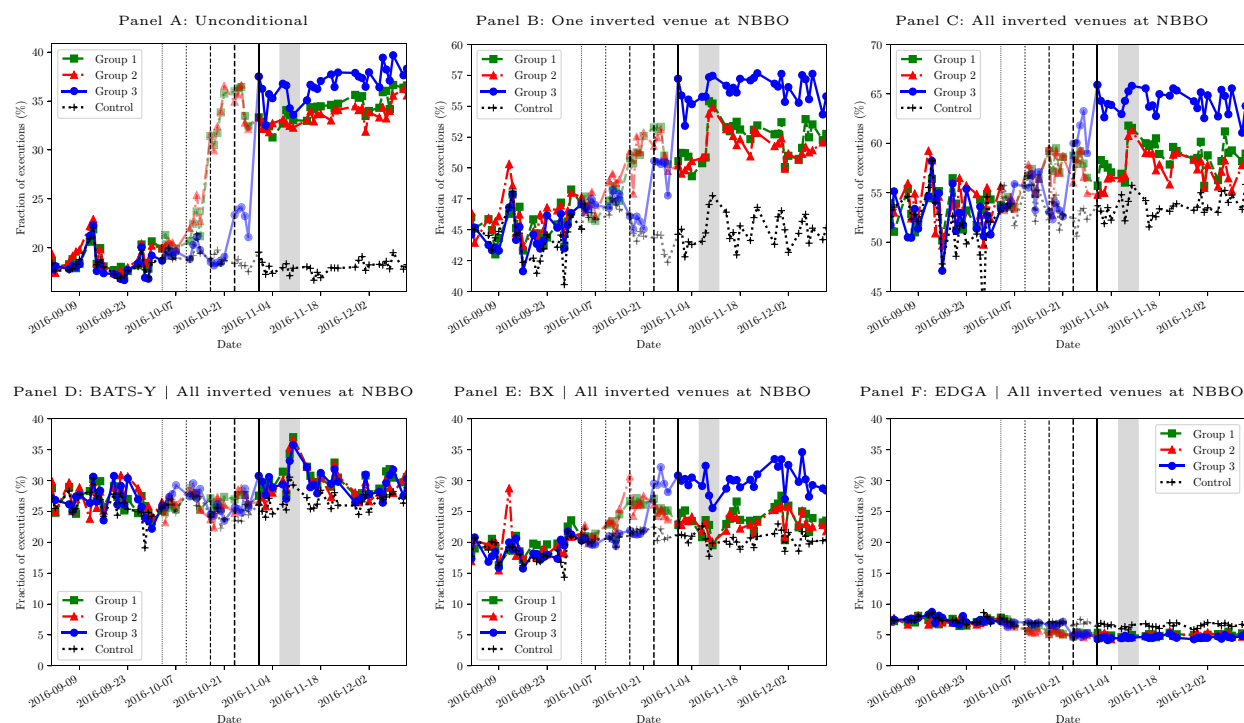


Fig. 5. Inverted venues fraction of executions. This figure shows the daily fraction of executions occurring on inverted venues unconditionally (Panel A), conditional on at least one inverted venue being at NBBO (Panel B), conditional on all inverted venues being at NBBO (Panel C), and the market share of specific inverted venues conditional on all inverted venues being at NBBO (Panels D, E, and F). Fractions of executions, in percentage terms, are presented separately for group one (green squares), group two (red triangles), group three (blue circles), and the control group (black crosses). The vertical lines indicate when stocks are added to the pilot groups. Five stocks were added to groups one and two at the first line, a further 95 stocks were added at the second line, and all remaining group one and two stocks are added at the third line. Five group three stocks are also added at the third line, a further 95 group three stocks were added at the fourth line, and the remaining group three stocks were added at the fifth line. The dark shaded band indicates the US election week. The sample period is September 1 to December 13, 2016. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

quality.²⁰ Instead, we use it to capture changes in trading activity, i.e., the net outcome of changes in order placement and execution on each venue during the pilot. Our previous results show that more marketable and nonmarketable limit orders are routed to inverted venues. The trade-to-order ratio is a good measure of the net benefit to liquidity providers, as it proxies for the probability of execution in equilibrium.²¹

We use the OLS regression specifications of Eqs. (6) and (7) to study how trading activity on NASDAQ and NASDAQ BX changes after the pilot implementation. The dependent variable is Trade-to-order_{*i,v,τ*}, the trade-to-order ratio for stock *i* on venue *v* (inverted or noninverted) on day *τ*. In addition to the same set of control variables used previously, we also include venue fixed effects to control for venue-specific differences.

Table 8 reports coefficient estimates of OLS regressions for inverted and noninverted venues. We find that the

trade-at rule increases the trade-to-order ratio on inverted venues by 0.88%. This increase is concentrated in high tick constraint stocks, for which the trade-to-order ratio increases by 1.03% more than for low tick constraint stocks. In contrast, the trade-at rule barely affects the ratio on noninverted venues. The trade rule does not yield any significant effect on the trade-to-order ratio. The quote rule increases the ratio significantly for noninverted venues but not for inverted venues. Moreover, unlike the trade-at rule, the impact of the quote rule is larger for low tick constraint stocks.²² The result on the quote rule effect on noninverted venues should be interpreted with caution. While it implies that the unconditional execution probability on

²⁰ Hasbrouck and Saar (2009) show that trade-to-order ratio fails to capture the fast cancellation of nonmarketable limit orders and hence is not a good measure for market quality.

²¹ Section I.f of the Online Appendix presents robustness tests using Rule 605 data that show that both the quote rule and the trade-at rule increase the execution rate of top-of-book orders on inverted venues.

²² We note that there is a potential bias due to missing observations. Unfortunately, SEC MIDAS does not provide data for all listed stocks on all exchanges. Overall, about 13% of observations are missing for BATS-Y, EDGA, and noninverted venues, while about 21% are missing for NASDAQ BX, and the proportion of missing values is similar for the low and high tick constraint subsamples. For the three inverted venues, between 12% and 37% of missing values correspond to days with no trades. We filter daily stock exchange observations for which there are fewer than 20 trades, as these would generate extreme values in our sample that could mechanically enhance our results, at the cost of excluding a significant fraction of our sample. Indeed, our results are robust to omitting the filter, but we obtain lower adjusted *R*². Results omitting the filter are presented in Table A.10 of the Online Appendix.

Table 8

Trade-to-order ratio. This table reports coefficients from OLS regressions of trade-to-order ratio, in %. The trade-to-order ratio is defined as trade volume divided by order volume. $\mathbf{1}_{[\text{s.t. trade-at}]}$, $\mathbf{1}_{[\text{s.t. trade}]}$, $\mathbf{1}_{[\text{s.t. quote}]}$ are dummy variables equal to one if the stock is subject to the rule at one point in the sample, and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. $\mathbf{1}_{[\text{tick const.}]}$ is a dummy variable equal to one if the stock has above-median tick constraint, and zero otherwise. The level of tick constraint is measured as the inverse of the time-weighted average spread during the pre-pilot period of September 2016. $\mathbf{1}_{[\text{post}]}$ is a dummy variable equal to one if the stock is in the post-treatment period, and zero otherwise. Dependent variables are at the stock-venue-day level, and results are presented for inverted fee venues and for non-inverted venues separately. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), venue fixed effects, index membership fixed effects, and industry fixed effects. t -statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. Sample descriptive statistics for the pre-pilot period are presented in Tables 1 and 4.

	Inverted		Noninverted	
	(1)	(2)	(3)	(4)
$\mathbf{1}_{[\text{s.t. trade-at}]} \times \mathbf{1}_{[\text{post}]}$	0.875*** (4.05)	0.224 (1.03)	0.209 (1.45)	0.329* (1.83)
$\mathbf{1}_{[\text{s.t. trade}]} \times \mathbf{1}_{[\text{post}]}$	0.257 (1.22)	0.435** (2.31)	0.166* (1.69)	0.146 (1.04)
$\mathbf{1}_{[\text{s.t. quote}]} \times \mathbf{1}_{[\text{post}]}$	-0.189 (-1.03)	0.121 (0.71)	0.498*** (4.83)	1.069*** (8.65)
$\mathbf{1}_{[\text{s.t. trade-at}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		1.033*** (3.00)		-0.181 (-0.81)
$\mathbf{1}_{[\text{s.t. trade}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		-0.292 (-0.87)		0.013 (0.07)
$\mathbf{1}_{[\text{s.t. quote}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		-0.341 (-1.16)		-0.881*** (-5.97)
Controls	Yes	Yes	Yes	Yes
Venue F.E.	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes
Adjusted R^2	0.197	0.212	0.320	0.322
N	223,621	223,621	443,169	443,169

noninverted venues is higher after the tick increase, this does not mean that the execution is faster; it could be that liquidity providers are taking more time before canceling orders. We show in Table A.9 of the Online Appendix that the higher trade-to-order ratio is driven at least in part by a decrease in canceled orders on noninverted venues.

Our analysis of the trade-to-order ratio indicates that the trade-at rule affects order activity on inverted venues more than on other lit venues. This is even more pronounced for high tick constraint stocks. Our results reflect the change in order placement strategy in response to the rule change. If those orders submitted and executed on inverted venues are informative, the increase in inverted trading would yield the positive impact on price discovery and thereby improve pricing efficiency. To test this hypothesis, we examine the contribution of marketable and non-marketable limit orders on inverted venues to price discovery.

6.3. Impact of order activity

In this section, we take a closer look at changes in the information contribution of order activity on inverted and noninverted venues following the pilot. Since the pilot has the most impact on NASDAQ BX, we focus on NASDAQ BX and its traditional counterpart, the main NASDAQ exchange, for this more granular analysis.²³ We obtain order-level data for both venues from NASDAQ TotalView-ITCH, which contains order-level messages with a nanosecond timestamp, from September 1 to December 13, 2016.

Following Fleming et al. (2017) and Brogaard et al. (2019), we extend the vector auto-regression approach of Hasbrouck (1991) to estimate the contribution of marketable and nonmarketable limit orders on NASDAQ and NASDAQ BX to the price discovery process. Specifically, we estimate the following VAR model with five lags:

$$\begin{aligned}
 r_t &= \sum_{i=1}^5 \alpha_i^0 r_{t-i} + \sum_{i=1}^5 \beta_i^{0,1} X_{t-i}^1 + \sum_{i=1}^5 \beta_i^{0,2} X_{t-i}^2 \\
 &\quad + \cdots + \sum_{i=1}^5 \beta_i^{0,10} X_{t-i}^{10} + \mu_t^0 \\
 X_t^1 &= \sum_{i=1}^5 \alpha_i^1 r_{t-i} + \sum_{i=1}^5 \beta_i^{1,1} X_{t-i}^1 + \sum_{i=1}^5 \beta_i^{1,2} X_{t-i}^2 \\
 &\quad + \cdots + \sum_{i=1}^5 \beta_i^{1,10} X_{t-i}^{10} + \mu_t^1 \\
 X_t^2 &= \sum_{i=1}^5 \alpha_i^2 r_{t-i} + \sum_{i=1}^5 \beta_i^{2,1} X_{t-i}^1 + \sum_{i=1}^5 \beta_i^{2,2} X_{t-i}^2 \\
 &\quad + \cdots + \sum_{i=1}^5 \beta_i^{2,10} X_{t-i}^{10} + \mu_t^2 \\
 &\quad \vdots \\
 X_t^{10} &= \sum_{i=1}^5 \alpha_i^{10} r_{t-i} + \sum_{i=1}^5 \beta_i^{10,1} X_{t-i}^1 + \sum_{i=1}^5 \beta_i^{10,2} X_{t-i}^2 \\
 &\quad + \cdots + \sum_{i=1}^5 \beta_i^{10,10} X_{t-i}^{10} + \mu_t^{10},
 \end{aligned} \tag{8}$$

where r_t is the midpoint log return between event times $t-1$ and t , and X_t^j are the ten order activity variables. The order activity variables are signed indicator variables that take the value -1 or 1 when the corresponding event occurs on event time t , and 0 otherwise. We keep track of five distinct event types occurring on NASDAQ and NASDAQ BX, for a total of ten event types: marketable order, nonmarketable limit order added at NBBO, non-marketable limit order added away from NBBO, nonmarketable limit order canceled at NBBO, and nonmarketable limit order canceled away from NBBO. Events related to marketable or nonmarketable limit buy orders are signed +1, while events

²³ In addition, the NASDAQ venues are the only two for which we can obtain order-level data. This provides a clean comparison because these two markets use the same technology and order types but differ in their fee structure.

triggered by marketable or non-marketable limit sell orders are signed -1 .

We estimate the model at the stock-day level, using all events between 9:45 a.m. ET and 3:45 p.m. ET. We exclude all stock-day observations for which there are less than five events of each type on each exchange. This requirement excludes 50,333 stock-day observations.²⁴ We then compute the orthogonalized cumulative impulse response function over 20 periods for each event type, corresponding to the average permanent price impact, in basis points, of the arrival of a marketable or nonmarketable limit buy order.

We use the OLS regression specifications of Eqs. (6) and (7) to study how orders on NASDAQ and NASDAQ BX affect price discovery after the pilot implementation. The dependent variables are $CIRF_{i,o,\tau}$ and $CIRF_{i,o,\tau} \times N_{i,o,\tau}$, where $CIRF_{i,o,\tau}$ is the cumulative impulse response function for event type o , in basis points, for stock i on day τ , and $N_{i,o,\tau}$ is the number of events of type o for stock i on day τ .

Estimation results are presented in Table 9. Panel A presents results for the impulse response function of individual orders. Specifically, we report results for marketable orders on NASDAQ (MO N), marketable orders on NASDAQ BX (MO BX), nonmarketable limit order arrivals at the NBBO on NASDAQ (L@BBO N), and nonmarketable limit order arrivals at the NBBO on NASDAQ BX (L@BBO BX), respectively.

We find that marketable orders on both exchanges contribute more to price discovery after the implementation of the quote rule and the trade-at rule. After the rule changes, marketable orders exert a larger impact on the return process. First looking at the impact of the trade-at rule, we find it is significant for both exchanges but larger on NASDAQ BX. On NASDAQ, the increase is $(0.028/0.27 \approx) 10\%$ compared to the prepilot level, while the increase on NASDAQ BX is $(0.024/0.09 \approx) 27\%$. Given our hypothesis that the improvement in market quality is driven by informed liquidity providers seeking queue priority on inverted venues, we are especially concerned with the informativeness of nonmarketable limit orders on inverted venues. The effect of the trade-at rule on the impact of nonmarketable limit orders on NASDAQ BX is a significant increase $(0.042/0.20 \approx) 21\%$, which supports our hypothesis. The impact is only significant for high tick constraint stocks. The trade-at rule also increases the impact of nonmarketable limit orders on NASDAQ; however, the increase of $(0.033/0.32 \approx) 10\%$ of the prepilot level is smaller compared to NASDAQ BX.

Looking at the impact of the quote rule on the impact of marketable orders, on NASDAQ the increase is $(0.079/0.27 \approx) 29\%$ compared to the prepilot level, while the increase on NASDAQ BX is $(0.087/0.09 \approx) 97\%$. An increase in the impact of marketable orders following the implementation of the quote rule is somewhat expected. Since the cost of crossing the spread increases after the

rule change, marketable orders signal stronger order flow information (more aggressiveness) and hence are more likely to move prices. While the contribution to price discovery of nonmarketable limit orders on NASDAQ is barely affected by the quote rule, there is a significant increase in the impact of nonmarketable limit orders on NASDAQ BX. The magnitude of the increase is $(0.121/0.20 \approx) 61\%$ of the prepilot level.

We also look at the aggregate daily impact of all events of a certain order type rather than the per order impact. In Panel B, we use the same dependent variables as in the previous set of regressions but scale them by the number of daily events of each type. We find that the aggregate impact of marketable orders on NASDAQ BX increases by about $(12.747/4.56 \approx) 280\%$ with the trade-at rule, much more than the per order increase of 27%, and the impact is significantly larger for tick constrained stocks. We also find that there is an increase in the aggregate informativeness of nonmarketable limit orders on both exchanges. On NASDAQ BX, the trade-at rule increases the aggregate impact of nonmarketable limit orders by $(124.786/159.77 \approx) 78\%$, much more than the per order increase of 21%. This suggests not only that each nonmarketable limit order on NASDAQ BX becomes more informative, but that there are also more of these informative nonmarketable limit orders being posted. This result is much stronger in the high tick constraint subsample. On NASDAQ, the increase is smaller at $(244.961/954.85 \approx) 26\%$.

The impact of the quote rule also yields interesting results. On the one hand, we find that it increases the information content revealed on NASDAQ BX by both marketable and nonmarketable limit orders. On the other hand, we find that it decreases the information content revealed on NASDAQ by both order types, despite the fact that individual orders are more informative.

Our order-level analysis suggests that the trade-at rule, which increases the market share of NASDAQ BX, is also associated with more informative marketable and nonmarketable limit orders on that venue. The improvement in informativeness is observed both at the order level and at the aggregate level. This is consistent with our hypothesis that informed traders are willing to pay a fee to offer sub-tick price improvement in exchange for a higher probability of execution. Since their orders are informative, this improves pricing efficiency. Our results are stronger for high tick constraint stocks because the demand for sub-tick price improvement is more important in these stocks.

6.4. Short selling

In this section, we examine whether the arrival of more informative nonmarketable limit orders on NASDAQ BX is associated with the arrival of informed traders. Using short selling as a proxy for informed trading (see, e.g., Boehmer et al., 2008; Comerton-Forde et al., 2016), we study changes in informed trading due to the implementation of the trade-at rule.

We obtain trade-level short selling data for NASDAQ BX from NASDAQ and for dark venues in aggregate from

²⁴ While Brogaard et al. (2019) use a minimum threshold of 20 events of each type, our sample is comprised of small stocks, many of which do not generate much activity, so a lower threshold avoids an overreduction of our sample. The additional estimation error is attenuated when averaging over our much larger cross-sectional sample.

Table 9

VAR estimation results. This table reports coefficients from OLS regressions of the cumulative impulse response function (CIRF), in basis points, of various market events. The market event are marketable limit orders (MO) and nonmarketable limit order arrivals at the NBBO (L@BBO) on NASDAQ (N) and NASDAQ BX (BX). Panel A presents results for the per-event CIRF, while Panel B presents results for the per-event CIRF scaled by the number of daily events. $\mathbf{1}_{[s.t. \text{trade-at}]}$, $\mathbf{1}_{[s.t. \text{trade}]}$, $\mathbf{1}_{[s.t. \text{quote}]}$ are dummy variables equal to one if the stock is subject to the rule at one point in the sample, and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. $\mathbf{1}_{[\text{tick const.}]}$ is a dummy variable equal to one if the stock has above-median tick constraint, and zero otherwise. The level of tick constraint is measured as the inverse of the time-weighted average spread during the pre-pilot period of September 2016. $\mathbf{1}_{[\text{post}]}$ is a dummy variable equal to one if the stock is in the posttreatment period, and zero otherwise. Controls include the daily VIX level, the log of the stock issue market cap (Log(Size)), the inverse of the share price (1/Price), index membership fixed effects, and industry fixed effects. *t*-statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, ** and, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program group three and control group. See the text for details on variable construction. Sample descriptive statistics for the pre-pilot period are presented in Tables 1 and 4.

Panel A: Cumulative IRF								
	MO N		MO BX		L@BBO N		L@BBO BX	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbf{1}_{[s.t. \text{trade-at}]} \times \mathbf{1}_{[\text{post}]}$	0.028** (2.16)	0.009 (0.49)	0.024** (2.61)	0.030** (2.47)	0.033** (2.38)	0.012 (0.65)	0.042*** (3.34)	0.016 (1.09)
$\mathbf{1}_{[s.t. \text{trade}]} \times \mathbf{1}_{[\text{post}]}$	0.001 (0.12)	0.013 (0.76)	0.003 (0.44)	0.007 (0.65)	0.016 (1.13)	0.006 (0.33)	−0.003 (−0.32)	−0.012 (−0.85)
$\mathbf{1}_{[s.t. \text{quote}]} \times \mathbf{1}_{[\text{post}]}$	0.079*** (7.57)	0.063*** (3.99)	0.087*** (13.46)	0.087*** (9.07)	−0.011 (−0.89)	−0.009 (−0.56)	0.121*** (11.62)	0.138*** (11.13)
$\mathbf{1}_{[s.t. \text{trade-at}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		0.025 (1.06)		−0.013 (−0.76)		0.034 (1.35)		0.040* (1.90)
$\mathbf{1}_{[s.t. \text{trade}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		−0.017 (−0.71)		−0.003 (−0.20)		0.019 (0.72)		0.016 (0.83)
$\mathbf{1}_{[s.t. \text{quote}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		0.016 (0.77)		−0.005 (−0.37)		−0.007 (−0.31)		−0.027 (−1.60)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.388	0.411	0.268	0.278	0.319	0.323	0.338	0.341
<i>N</i>	103,806	103,806	103,806	103,806	103,806	103,806	103,806	103,806
Panel B: Cumulative IRF $\times N$								
	MO N		MO BX		L@BBO N		L@BBO BX	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbf{1}_{[s.t. \text{trade-at}]} \times \mathbf{1}_{[\text{post}]}$	13.722* (1.87)	−1.603 (−0.17)	12.747*** (5.60)	4.053*** (2.69)	244.961*** (4.93)	93.155* (1.96)	124.786*** (5.64)	45.854* (1.91)
$\mathbf{1}_{[s.t. \text{trade}]} \times \mathbf{1}_{[\text{post}]}$	2.483 (0.40)	−6.342 (−0.83)	0.356 (0.27)	−0.162 (−0.11)	10.673 (0.25)	−30.141 (−0.63)	−13.128 (−0.93)	−25.804 (−1.20)
$\mathbf{1}_{[s.t. \text{quote}]} \times \mathbf{1}_{[\text{post}]}$	−22.082*** (−3.63)	−6.379 (−0.74)	14.673*** (11.48)	10.082*** (7.85)	−312.277*** (−8.52)	−234.405*** (−5.72)	113.907*** (7.17)	152.783*** (8.00)
$\mathbf{1}_{[s.t. \text{trade-at}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		26.491** (2.03)		14.325*** (3.96)		261.867*** (3.44)		132.617*** (3.72)
$\mathbf{1}_{[s.t. \text{trade}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		12.506 (1.15)		1.187 (0.48)		59.751 (0.78)		20.163 (0.71)
$\mathbf{1}_{[s.t. \text{quote}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		−23.713** (−2.21)		7.557*** (4.02)		−104.219 (−1.67)		−55.366** (−2.47)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.045	0.046	0.136	0.157	0.076	0.092	0.082	0.111
<i>N</i>	103,806	103,806	103,806	103,806	103,806	103,806	103,806	103,806

FINRA.²⁵ We exclude short-exempt sales, as they represent a very small fraction of short selling, and match the per stock daily dollar volume sold short with TRTH to compute the share of aggregate dollar volume that is sold short on NASDAQ BX and dark venues, respectively.

We use the OLS regression specifications of Eqs. (6) and (7) to study changes in $\text{Short rate}_{i,\tau}$, the short selling rate

on NASDAQ BX (Short BX) or on dark venues (Short dark), in percentage terms, for stock *i* at time τ . We define the short selling rate as the total dollar volume sold short, divided by the total dollar volume across all venues.

Results presented in Table 10 indicate that the trade-at rule significantly increases short selling on NASDAQ BX by 1.18% of the daily consolidated volume. This is economically significant considering the mean pre-pilot level of short selling on NASDAQ BX is 1.07%. In the subsample analysis, we find that high tick constraint stocks

²⁵ NASDAQ makes freely available this short sale trade report for NASDAQ BX but not for the main NASDAQ exchange.

Table 10

Short selling. This table reports coefficients from OLS regressions of short selling rate, in %. Dependent variables are presented for NASDAQ BX (Short BX) and dark venues (Short dark). The short selling rate is defined as the total dollar volume sold short, divided by the total dollar volume sold across all venues. $\mathbf{1}_{[\text{s.t. trade-at}]}$, $\mathbf{1}_{[\text{s.t. trade}]}$, $\mathbf{1}_{[\text{s.t. quote}]}$ are dummy variables equal to one if the stock is subject to the rule at one point in the sample, and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. $\mathbf{1}_{[\text{tick const.}]}$ is a dummy variable equal to one if the stock has above-median tick constraint, and zero otherwise. The level of tick constraint is measured as the inverse of the time-weighted average spread during the pre-pilot period of September 2016. $\mathbf{1}_{[\text{post}]}$ is a dummy variable equal to one if the stock is in the posttreatment period, and zero otherwise. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), index membership fixed effects, and industry fixed effects. t -statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, ** and, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. Sample descriptive statistics for the pre-pilot period are presented in Tables 1 and 4.

	Short BX		Short dark	
	(1)	(2)	(3)	(4)
$\mathbf{1}_{[\text{s.t. trade-at}]} \times \mathbf{1}_{[\text{post}]}$	1.181*** (13.58)	0.698*** (7.62)	−7.436*** (−22.97)	−6.279*** (−14.46)
$\mathbf{1}_{[\text{s.t. trade}]} \times \mathbf{1}_{[\text{post}]}$	−0.046 (−0.81)	0.022 (0.29)	−0.341 (−1.23)	−0.448 (−1.11)
$\mathbf{1}_{[\text{s.t. quote}]} \times \mathbf{1}_{[\text{post}]}$	1.311*** (27.10)	0.940*** (14.86)	2.157*** (8.25)	0.538 (1.44)
$\mathbf{1}_{[\text{s.t. trade-at}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		0.861*** (6.42)		−2.352*** (−4.18)
$\mathbf{1}_{[\text{s.t. trade}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		−0.105 (−1.05)		0.297 (0.55)
$\mathbf{1}_{[\text{s.t. quote}]} \times \mathbf{1}_{[\text{post}]} \times \mathbf{1}_{[\text{tick const.}]}$		0.719*** (8.70)		3.114*** (6.84)
Controls	Yes	Yes	Yes	Yes
Index F.E.	Yes	Yes	Yes	Yes
Industry F.E.	Yes	Yes	Yes	Yes
Adjusted R^2	0.278	0.299	0.110	0.113
N	150,832	150,832	150,832	150,832

experience a more substantial increase than low tick constraint stocks. The quote rule also increases short selling on NASDAQ BX by 1.31%, which is of similar magnitude to the increase caused by the trade-at rule.

The impact of the pilot on short selling is different for dark venues. Short selling on dark venues decreases by 7.44% when the trade-at rule is implemented. However, it increases by 2.16% when the quote rule is implemented. This implies that when the tick size increases, short sellers route their orders to alternative venues with sub-tick trading features, e.g., inverted and dark venues. When the trade-at rule is enforced, short sellers leave dark venues, and some of them migrate to inverted venues.

To assess the importance of short selling by liquidity providers, we classify short trades on NASDAQ BX by the type of order, marketable or nonmarketable limit orders, used by the short seller. We follow Comerton-Forde et al. (2016) and merge trades from the short sale report with trades in the ITCH feed by matching on price, volume, and timestamp with second precision. If more than one trade in ITCH is a match for a given trade in the short sale report, we flag the first trade in ITCH that matches. For most trades, we can identify if it is a buyer- or seller-initiated trade directly from the ITCH order-level data. However, for trades against hidden orders, the ITCH feed does not identify the aggressor. We sign these trades with the Lee and Ready (1991) algorithm using the prevailing midquote one nanosecond before the execution. We then exclude short sale trades that are left unsigned (i.e., trades at the

midquote). For each stock-day observation, we compute the share of short sales by liquidity providers as the ratio of the number of short sales on NASDAQ BX for which the liquidity provider is the seller, divided by the total number of short sales on NASDAQ BX. We drop stock-day observations for which there are no short trades or for which we are unable to sign at least one short trade.

We use the same OLS regression specification to examine the effect of the pilot policy on short selling by liquidity providers and present results in Table 11. We find that the trade-at rule significantly increases the ratio of short selling by liquidity providers on NASDAQ BX by 3.35% more for high than low tick constraint stocks, where the latter group exhibit a negative (and insignificant) effect. The quote rule increases the ratio of liquidity providing short sales for both subsamples, but this effect is more pronounced for low tick constraint stocks.

Combining the impact of the pilot policy on short sales, and on liquidity providing short sales in particular, we find that more informed traders move to NASDAQ BX following the policy change. When the tick size increases to one nickel, short sellers, and especially liquidity providing short sellers, become more active on inverted venues. The trade-at rule pushes even more short sellers to NASDAQ BX and leads to a larger ratio of liquidity providing short sales for stocks that are more likely to be tick constrained. These results show that inverted venues, such as NASDAQ BX, attract informed liquidity providers who wish to offer sub-tick price improvement. This results in more informative

Table 11

Short selling by liquidity providers. This table reports coefficients from OLS regressions of the share of short sales by liquidity providers, in %, on rule dummies ($1_{\text{trade-at}}$, 1_{trade} , and 1_{quote}) and controls. The share of short sales by liquidity providers is defined as the number of short sales on NASDAQ BX for which the liquidity provider is the seller, divided by the total number of short sales on NASDAQ BX. $1_{\text{s.t. trade-at}}$, $1_{\text{s.t. trade}}$, $1_{\text{s.t. quote}}$ are dummy variables equal to one if the stock is subject to the rule at one point in the sample, and zero otherwise. The trade-at rule applies to group three only, the trade rule applies to groups two and three, and the quote rule applies to all pilot tests groups. $1_{\text{tick const.}}$ is a dummy variable equal to one if the stock has above-median tick constraint, and zero otherwise. The level of tick constraint is measured as the inverse of the time-weighted average spread during the prepilot period of September 2016. 1_{post} is a dummy variable equal to one if the stock is in the posttreatment period, and zero otherwise. Controls include the daily VIX level, the log of the stock issue market cap ($\text{Log}(\text{Size})$), the inverse of the share price ($1/\text{Price}$), index membership fixed effects, and industry fixed effects. t -statistics from robust standard errors clustered by date and stock are presented in parenthesis, and *, ** and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is September 1, 2016, to December 13, 2016, and include stocks from the Tick Size Pilot Program groups and control group. Sample descriptive statistics for the prepilot period are presented in Tables 1 and 4.

	BX limit short rate	
	(1)	x(2)
$1_{\text{s.t. trade-at}} \times 1_{\text{post}}$	0.160 (0.17)	−1.780 (−1.30)
$1_{\text{s.t. trade}} \times 1_{\text{post}}$	1.208 (1.55)	−0.461 (−0.35)
$1_{\text{s.t. quote}} \times 1_{\text{post}}$	5.217*** (8.60)	6.927*** (6.40)
$1_{\text{s.t. trade-at}} \times 1_{\text{post}} \times 1_{\text{tick const.}}$		3.345** (2.09)
$1_{\text{s.t. trade}} \times 1_{\text{post}} \times 1_{\text{tick const.}}$		2.622 (1.59)
$1_{\text{s.t. quote}} \times 1_{\text{post}} \times 1_{\text{tick const.}}$		−2.838** (−2.05)
Controls	Yes	Yes
Index F.E.	Yes	Yes
Industry F.E.	Yes	Yes
Adjusted R ²	0.024	0.026
N	119,949	119,949

order activity on NASDAQ BX, leading to better pricing efficiency and liquidity for the overall market.

7. Conclusion

Make-take fee models are controversial. Our study is timely, as the impact of these fee models is receiving increased market and regulatory attention. In December 2014, the NYSE proposed a “grand bargain” that included a ban on make-take and inverted fees. In response, BATS proposed a significant reduction in access fees but opposed a ban on rebates. NASDAQ conducted a limited pilot study to test the premise that high access fees discourage the use of displayed nonmarketable limit orders. In July 2016, the SEC Market Structure Advisory Committee recommended that the SEC undertake an Access Fee Pilot. In March 2017, Acting SEC Chair Piwowar directed the commission to launch this pilot plan.

Utilizing the SEC Tick Size Pilot as a natural experiment, we carefully evaluate the effect and identify one channel through which inverted fee models affect market quality. We find increases in trading on inverted venues improves

pricing efficiency but only in stocks where the minimum tick size is binding. We show that the improvement occurs because inverted venues enable traders to trade in sub-tick prices, which allows additional information to be incorporated into prices. Using order-level data on NASDAQ and its inverted venue NASDAQ BX, we find that the information content of nonmarketable limit orders on NASDAQ BX increases significantly after the trade-at rule implementation. We also find, using short sale trade reports, that informed traders become more active on NASDAQ BX after the rule is implemented. In all our results, the finding is stronger for stocks that are tick constrained.

Our results provide support for the view that inverted pricing models can improve pricing efficiency by increasing competition for liquidity provision (Biais et al., 2010). We show that by allowing sub-tick trading, inverted fee models facilitate the price discovery process when the tick size is binding. As the sub-tick pricing grid offered by inverted fee venues is a feature they share with many dark venues, our results also suggest that this sub-tick pricing grid can partially account for the positive effects of dark trading shown by Buti et al. (2015).

However, our results do not unambiguously support inverted fee venues. The benefits accrue only in stocks where the minimum tick size is binding—suggesting an alternative policy approach may be to focus on finding optimal tick sizes more dynamically. This is a view we share with O'Hara et al. (2019). In addition, there are other significant drawbacks to the competition on exchange fee models shown in the literature. They make the market more complex and more fragmented. New order types have proliferated as exchanges seek to provide opportunities for traders to capture rebates and avoid fees, further increasing complexity. They also render quotes less informative because they are displayed without adjusting for fees and exacerbate agency conflicts between brokers and their clients. All of these factors should be carefully considered by the SEC when designing the next pilot—the Access Fee Pilot.

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